# A fast kernel for $D_n(G) = G \cap (1 + \Delta^n)$: profile, variant A, variant B

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The starting point is the kernel of the reference implementation `dimsub.py` (method A in numpy; it lives in a separate, so far private repository). This document covers its profile, a port to C (variant A), a different way of computing (variant B) with a proof of correctness, the comparison, timings and recommendations.

Notation: $G$ is a finite $p$-group, $x_1, \dots, x_d$ are its generators, $\Delta \subset \mathbb{Z}[G]$ is the augmentation ideal, $F$ is the free group on $x_1 \dots x_d$, $\mathfrak{f} \subset \mathbb{Z}[F]$ is its augmentation ideal, $A_n = \mathbb{Z}\langle a_1, \dots, a_d\rangle/(\text{degree} \geq n)$ is the truncated free associative algebra, $A_n^+$ is its part without constant term, $M = \operatorname{rk} A_n^+ = d + d^2 + \cdots + d^{n-1}$.

## 0. Summary

- **Variant B is correct** (proof in §3) and **confirmed by comparison**: 0 differences from `dimsub.py` on every group for which answers exist (§5), and full agreement with an independent implementation of the same method, in both directions.
- **Speed.** On groups of order $3^7$: numpy 80–330 s, A in C 7–22 s (×11–15), B 0.2–1.4 s (×230–380, including Python start-up). Variant B went through all 1202 candidates of order $3^7$ in 157 s on three threads (numpy: about 100 CPU

hours).

- **Most importantly, B does not depend on** $|G|$**.** The Bartholdi–Mikhailov group of order $3^{494}$ ($d = 7$): $\mathbb{Z}[G]/\Delta^4$ in 15 s modulo $3^{19}$, the exact HNF over $\mathbb{Z}$ in 510 s, with the same indices. Rips-type 2-groups of orders $2^{40}$–$2^{107}$: seconds per group.
- **The limit of B** is $M = \sum_{k<n} d^k$, which grows like $d^{n-1}$. For $d = 6$, $n = 7$ (the plateau question of [IMP26, §3.3]) $M = 55\,986$, and dense linear algebra does not work there (estimate in §6.3).
- **FLINT's `nmod_mat_howell_form`** is slower than even numpy on these matrices (§2); our own Howell form for the local ring $\mathbb{Z}/p^E$ is 10–15 times faster than numpy.

## 1. Profile of the current kernel (method A, numpy)

Method A: $\Delta \cong \mathbb{Z}^{|G|-1}$, $\Delta^n = \mathrm{span}\{b(x_i - 1)\}$ over a basis $b$ of the lattice $\Delta^{n-1}$, Howell form modulo $p^E$ ($E = 19$ for $p = 3$, 30 for $p = 2$), a loop over columns in Python, rows processed in vectorised form.

Measured with `scripts/profile_dimsub.py` on the test machine, one process under `nice -n 10`. The machine (16 threads) is shared with unrelated CI jobs; the load average during the measurements was 15–29, so the numbers are noisy by tens of percent.

| group | $d$ | matrix per level | howell per level $n = 2, 3, 4, 5$, s | test $g \in D_n$, s | total, s |
|---|---|---|---|---|---|
| 2187 #171 | 2 | $4372 \times 2186$ | 4.9 / 14.3 / 22.2 / 28.9 | 1.8 | 73.5 |
| 2187 #4721 | 3 | $6558 \times 2186$ | 62.3 / 37.4 / 67.7 / 76.2 | 8.0 | 253.6 |
| 2187 #9066 | 4 | $8744 \times 2186$ | 74.4 / 47.2 / 86.1 / 91.2 | 6.7 | 307.9 |
| 512 #864 | 2 | $1022 \times 511$ | 0.34 / 0.24 / 0.39 | 0.06 | 1.1 |
| 512 #405386 | 4 | $2044 \times 511$ | 0.94 / 0.81 / 1.31 | 0.08 | 3.3 |

- **All the time goes into the Howell form**, 95–98 %. Building the blocks $b \cdot (x_i - 1)$ takes a fraction of a second, the membership test a few percent.
- The matrix at every level has the same size, $d(|G| - 1) \times (|G| - 1)$, and the rank is always $|G| - 1$. The cost is $\sim d|G|^3/2$ operations per level. The growth of the time with $n$ comes from the $p$-torsion of $\Delta/\Delta^n$: the rows $p^{E-v}\cdot$(pivot) return to the pool.
- In the original run over the sample of order 2187 (14 processes in parallel) the median was 278 s and the maximum 659 s. A single process is 2–4 times faster: under full load numpy is memory-bound.
- The order $|G|^3$ is the main limit: $2^{10}$ costs 8 times more per level than 512, and a group of order $2^{38}$ cannot be enumerated at all.

## 2. Variant A: a port to C

`c/dimsub_a.c` and `c/hw.h`: the same algorithm, the same $E$, the same output as `dimsub.analyse`, in fast mode and with `--full`. The check "$\Delta^n$ is a right ideal" from `--full` is not ported.

- **Own Howell form** (the default). The ring $\mathbb{Z}/p^E$ is local, so the pivot is the row with the least $p$-adic valuation in the column, and it divides the others: one subtraction per entry, no `xgcd` on pairs of rows. Rows are stored as `uint32` ($p^E < 2^{31}$), `row -= f·piv` is computed in `uint64` with reduction through `double` ($q = \lfloor x/D \rfloor$ with error $\le 1$ for $x < 2^{63}$, one correction), without division. The loop vectorises (`-O3 -march=native`; the test machine has AVX-512DQ).
- **FLINT** (`--flint`): `nmod_mat_howell_form` from FLINT 3.3 (conda-forge `libflint`). It is general for $\mathbb{Z}/N$: for each pair of rows an `xgcd` and a unimodular $2 \times 2$ transformation; moreover, it needs at least as many rows as columns. On our matrices it is **slower than numpy** (table in §4).
- Delayed reduction (accumulating several `row -= f·piv` without taking the remainder) does not fit into 64 bits at $p^E \approx 2^{31}$: each product is already $\approx 2^{62}$. It makes sense only for smaller $E$ (§6).

## 3. Variant B: $\mathbb{Z}[G]/\Delta^n$ as a quotient of the truncated free algebra

### 3.1. Statement

Let $G = F/N$, $\pi : \mathbb{Z}[F] \to \mathbb{Z}[G]$, $\mathfrak{r} = \ker \pi$, and let $\mu : \mathbb{Z}[F] \to A_n$ be the Magnus ring homomorphism $x_i \mapsto 1 + a_i$ (then $x_i^{-1} \mapsto \sum_{k<n}(-a_i)^k$). Let $R_0 \subset F$ normally generate $N$.

**Theorem.** $\mathbb{Z}[G]/\Delta^n \cong A_n/J_n$, where $J_n = \operatorname{span}_{\mathbb{Z}}\{\, u\,(\mu(r)-1)\,v \,:\, r \in R_0;\ u, v \text{ monomials}\}$ is the two-sided ideal generated by the elements $\mu(r)-1$. The isomorphism sends $g$ to $\mu(w_g)$ for any word $w_g$ representing $g$. Therefore

$$g \in D_n(G) \iff \mu(w_g) - 1 \in J_n.$$

Moreover, $A_n^+/J_n \cong \Delta/\Delta^n$ is a finite $p$-group, and $\operatorname{rk} A_n^+ = M$ does not depend on $|G|$.

### 3.2. Proof

**(a) The truncated Magnus isomorphism:** $\mathbb{Z}[F]/\mathfrak{f}^n \cong A_n$**.** $\mu$ is well defined: $1 + a_i$ is invertible in $A_n$ (the inverse is $\sum_{k<n}(-a_i)^k$, since $a_i^n = 0$), and $F$ is free, so $x_i \mapsto 1 + a_i$ extends to a homomorphism $F \to A_n^\times$, and linearly to a ring homomorphism $\mathbb{Z}[F] \to A_n$. The constant term of $\mu(f)$ equals the augmentation $\varepsilon(f)$: both maps are ring homomorphisms to $\mathbb{Z}$ that agree on the $x_i$. Hence $\mu(\mathfrak{f}) \subseteq A_n^+$ and $\mu(\mathfrak{f}^n) \subseteq (A_n^+)^n = 0$, which gives $\bar\mu : \mathbb{Z}[F]/\mathfrak{f}^n \to A_n$. Conversely, by the universal property of the free associative algebra there is a ring homomorphism $\varphi : \mathbb{Z}\langle a\rangle \to \mathbb{Z}[F]/\mathfrak{f}^n$, $a_i \mapsto x_i - 1$. A monomial of degree $\geq n$ goes to a product of $n$ elements of $\mathfrak{f}$, that is, to 0, so $\varphi$ factors through $A_n$. Next, $\bar\mu\varphi(a_i) = a_i$, hence $\bar\mu\varphi = \mathrm{id}$. And $\varphi\bar\mu(x_i) = x_i$, hence $\varphi\bar\mu$ is the identity on $F$ (on $x_i^{-1}$ because a ring homomorphism preserves inverses), and $F$ additively generates $\mathbb{Z}[F]$. □

Magnus's deep theorem ($\mu^{-1}(\deg \geq n) = \mathfrak{f}^n$ in $\mathbb{Z}\langle\langle a\rangle\rangle$) is not needed here: for the truncation the statement is elementary.

**(b)** $\ker\pi = \mathfrak{r}$ **is the two-sided ideal** $I$ **generated by** $\{r-1 : r \in R_0\}$**.** The inclusion $I \subseteq \ker\pi$ is obvious. For $n \in N$ we have $n - 1 \in I$, since $wrw^{-1} - 1 = w(r-1)w^{-1}$, $r^{-1} - 1 = -r^{-1}(r-1)$ and $ab - 1 = a(b-1) + (a-1)$: induction on the length of $n$ as a product of conjugates of $r^{\pm1}$. Finally, $\ker\pi \subseteq \mathbb{Z}[F](N-1)$: if $\sum c_f f \in \ker\pi$, then the sum of the coefficients over each coset $tN$ is zero, and $\sum_{f\in tN} c_f f = t\sum c_f(t^{-1}f - 1)$ with $t^{-1}f \in N$. □

(b′) If $S$ generates $N$ **as a subgroup** (for example, $S$ the Schreier generators), then $\ker\pi = \sum_{s\in S}(s-1)\mathbb{Z}[F]$, and a **right** ideal suffices: $ns - 1 = (n-1)s + (s-1)$, $s^{-1} - 1 = -(s-1)s^{-1}$, and $\ker\pi = (N-1)\mathbb{Z}[F]$ by the same computation with right cosets.

**(c) Putting it together.** $\pi(\mathfrak{f}) = \Delta$ (the image of the augmentation ideal under a surjection of group rings), so $\pi(\mathfrak{f}^n) = \Delta^n$ and $\mathbb{Z}[G]/\Delta^n \cong \mathbb{Z}[F]/(\mathfrak{r} + \mathfrak{f}^n)$. Under $\bar\mu$ the submodule $(\mathfrak{r} + \mathfrak{f}^n)/\mathfrak{f}^n$ goes to $\mu(\mathfrak{r})$. The image of an ideal under a surjective ring homomorphism is the ideal generated by the images of the generators, that is, $\mu(\mathfrak{r}) = J_n$. As an abelian group it is spanned by $u\,\mu(r-1)\,v$ with monomials $u, v$, because $A_n$ is additively generated by monomials. Finally, $g - 1 \in \Delta^n \iff w_g - 1 \in \mathfrak{r} + \mathfrak{f}^n \iff \mu(w_g) - 1 \in J_n$. □

**(d) Finiteness and exactness modulo** $p^E$**.** $\varepsilon(J_n) = 0$, so $A_n = \mathbb{Z} \oplus A_n^+$ and $A_n^+/J_n \cong \Delta/\Delta^n$. Multiplication gives an epimorphism $(\Delta/\Delta^2)^{\otimes k} \to \Delta^k/\Delta^{k+1}$, and $\Delta/\Delta^2 \cong G_{ab}$ is a finite $p$-group. Hence all $\Delta^k/\Delta^{k+1}$ and $\Delta/\Delta^n$ itself are finite $p$-groups. Therefore **the exactness certificate of dimsub.py carries over verbatim**. Non-membership $\mu(w_g) - 1 \notin J_n + p^E A_n^+$ is always exact. Membership is exact if $p^{E-1}A_n^+ \subseteq J_n + p^E A_n^+$: then the exponent of $A_n^+/J_n$ divides $p^{E-1}$ and $p^E A_n^+ \subseteq J_n$. Moreover, $\log_p |A_n^+/(J_n + p^E A_n^+)| = \log_p |\Delta/(\Delta^n + p^E\Delta)|$, so the field `index_logp` must agree with dimsub.py. In the comparison it does.

### 3.3. Where the relations come from

**From the multiplication table** (`py/magnus.py`; the input is the same JSON line as for dimsub.py). A breadth-first search on the Cayley graph gives a spanning tree and words $w_g$; each edge $(g, x_i)$ gives

$$\rho_{g,i} = \mu(w_g)(1 + a_i) - \mu(w_{gx_i}) = \big(\mu(s_{g,i}) - 1\big)\,\mu(w_{gx_i}), \qquad s_{g,i} = w_g x_i w_{gx_i}^{-1}.$$

The elements $s_{g,i}$ are the Schreier generators of the kernel $N$ (they generate it as a subgroup), and $\mu(w_{gx_i})$ is invertible. By (b′) $J_n = \operatorname{span}\{\rho_{g,i}\,v\}$: closure under multiplication **on the right** by $a_j$ suffices, and that is a shift of coordinates. No inverses are needed; all $\mu(w_g)$ are computed along the tree in $O(|G| \cdot M)$. There are $(d-1)|G| + 1$ relations, but only $M$ columns, and the Howell form is accumulated in chunks.

**From a pc-presentation, without enumerating the group** (`gap/export_pres.g` → `py/magnus_pres.py`). GAP builds a pc-sequence refining the lower central series. Layer $k$ is a modulo pcgs of $\gamma_k/\gamma_{k+1}$, collected by sifting the commutators $[b, x_j]$ ($b$ an element of layer $k-1$) and their $p$-th powers, so that an element of layer $k$ has a word $W \in \gamma_k(F)$. The relations are: power relations $W_i^p = \prod W_j^{e_j}$, commutator relations $[W_j, W_i] = \prod W_k^{e_k}$, and $x_j = \prod W_k^{e_k}$. This is a presentation of $G$ on $x_1..x_d$ by Tietze transformations: the maps $G = \langle s \mid \mathrm{pc}\rangle \to \Gamma$, $s_i \mapsto W_i$, and $\Gamma \to G$ are mutually inverse, which is checked directly on the $s_i$ and on the $x_j$.

*Cut-off by weight.* $\mu(\gamma_k(F)) \subseteq 1 + A^{\geq k}$, by induction on $k$ from the identity $[x,y]-1 = x^{-1}y^{-1}\big((x-1)(y-1)-(y-1)(x-1)\big)$. Hence a relation from $\gamma_n(F)$ gives $\mu(r)-1=0$ in $A_n$ and need not be written out. We drop the power relations of layers $\geq n$ and the commutator relations with $k_i + k_j \geq n$: their right-hand sides lie in $\gamma_{k_i+k_j}(G)$, that is, in layers $\geq k_i + k_j$. For the same reason the words of elements of layers $\geq n$ are replaced by the identity. Without this, the words of the deep layers of the group of order $3^{494}$ do not fit into 8 GB.

### 3.4. All levels at once

$J_n$ is the image of $J_N$ ($n < N$) under dropping the degrees $\geq n$. If the columns are ordered by increasing degree, then the rows of the Howell form of $J_N$ with pivot column of degree $< n$, truncated to degrees $< n$, form a Howell form of $J_n$. Indeed, an element of the projection that is zero before column $c$ lifts to $J_N$ with the same first coordinates and decomposes over the rows with pivot $\geq c$. The same holds for the HNF over $\mathbb{Z}$. `magnus_pres.py` computes $J_{n_{\max}}$ once.

### 3.5. Testing a candidate subgroup $K$ (sieve)

If $K \subseteq D_{n-1}$ is an elementary abelian $p$-subgroup with independent generators $g_1..g_r$, then $g \mapsto (g-1) \bmod \Delta^n$ is a homomorphism on $K$: $gh-1 = (g-1)+(h-1)+(g-1)(h-1)$ and $(g-1)(h-1) \in \Delta^{2n-2} \subseteq \Delta^n$. Hence $D_n \cap K$ is the kernel of $c \mapsto \sum c_i v_i$ modulo $J_n$, $v_i = \mu(w_{g_i}) - 1$. This is a single Howell form of the matrix $[J_n \mid 0;\ V \mid I;\ 0 \mid pI]$: the rows with pivot in the last $r$ columns generate the kernel (`magnus_pres.py --kernel`). Only the words for $g_1..g_r$ are needed, not for all elements of the group.

### 3.6. Where B loses

Variant B has $M = \sum_{k=1}^{n-1} d^k$ columns, variant A has $|G|-1$. In the sample of order $3^7$ ($d \leq 4$, $n \leq 5$) $M \leq 340$ versus 2186; for the surviving groups of order 512 ($d \leq 5$, $n \leq 4$) $M \leq 155$ versus 511. But for large $d$ and large $n$ the rank of the free algebra overtakes $|G|$: for $C_2^6$ at $n = 6$ $M = 9330$, for groups of order $2^{10}$ with $d = 6$ at $n = 5$ $M = 1554 > 1023$. Hence two rules.

- In the full comparison mode (all levels up to $n = 6$) B stopped on groups with $M > 500$ and was compared on the common prefix of levels (the field `truncated`; the number of such groups is in §5).
- For each group, choose B if $M < |G|$, and A otherwise.

The real remedy is to store not the dense $A_n$ but only the "standard" monomials. With an ordering compatible with multiplication (deg-lex), a monomial that leads a row of $J$ with an invertible coefficient stays leading with an invertible coefficient in all multiples $u\,w\,v$ as well, so it is eliminated together with them. What remains is an order ideal of monomials, of size at least $\dim_{\mathbb{F}_p} \Delta_{\mathbb{F}_p}/\Delta^n_{\mathbb{F}_p}$ (§6.4). This is the nilpotent quotient algorithm for an associative algebra, but over the ring $\mathbb{Z}/p^E$ rather than over a field as in ModIsom. It is not part of the prototype (§6).

## 4. Timings

One group, one process, `nice -n 10`, the test machine under CI load (`scripts/bench.sh`), time in seconds. About 0.15 s of "B (Python)" is interpreter start-up.

| group | $d$ | numpy A (dimsub.py) | A in C | A in C, FLINT | B (Python) | A C / numpy | B / numpy |
|---|---|---|---|---|---|---|---|
| 2187 #171 | 2 | 79.4 | 7.0 | 111.2 | 0.21 | ×11 | ×380 |
| 2187 #4721 | 3 | 263.1 | 14.8 | 248.5 | 0.75 | ×18 | ×350 |
| 2187 #9066 | 4 | 332.4 | 22.4 | 272.6 | 1.42 | ×15 | ×230 |
| 512 #864 | 2 | 1.21 | 0.11 | 1.01 | 0.21 | ×11 | ×6 |
| 512 #405386 | 4 | 2.51 | 0.21 | 2.71 | 0.31 | ×12 | ×8 |

Whole sets (three threads each, wall time):

| set | groups | B (Python) | A in C | numpy in the original run, CPU |
|---|---|---|---|---|
| orders 8–625, full mode (8 sets) | 2 762 | 509 s | 9.1 s | — |
| 729, all | 504 | 9.2 s | 25.7 s | 13 780 s |
| 256, class ≥ 3 | 24 328 | 1 196 s | 131 s | 44 370 s |

| set | groups | B (Python) | A in C | numpy in the original run, CPU |
|---|---|---|---|---|
| 512, survivors of the sieve | 2 792 | 88 s | 65 s | 23 732 s |
| 2187, sample | 49 | 7.4 s | 248 s | 17 198 s |
| 2187, all candidates | 1 202 | 157 s | 4 905 s | ≈ 420 000 s (estimate, ≈ 117 h) |

At small orders (256, 512) A in C is faster than the Python variant B: there $M$ is comparable to $|G|$ (for groups of order 256 with $d = 6$ at $n = 5$ $M = 1554 > 255$), and Python has a fixed overhead per group. At order 2187 B is 30 times faster than A in C, and hundreds of times faster than numpy.

Large groups, only B from a presentation:

| group | $\|G\|$ | $d$ | $n$ | relations after the cut-off | $M$ | time |
|---|---|---|---|---|---|---|
| Bartholdi–Mikhailov [BM23] | $3^{494}$ | 7 | 4 | 6045 | 399 | 15 s mod $3^{19}$; 510 s exact (HNF over $\mathbb{Z}$, |
| Rips-type family (`gap/xcheck_fam.g`) | $2^{40}$–$2^{107}$ | 4 | 5 | 380–2330 | 340 | 1.7–9 s per group |

For the group of order $3^{494}$ the indices $\log_3 |\Delta/\Delta^n| = 59, 321, 1430$ ($n = 2, 3, 4$) agreed modulo $3^{19}$ and over $\mathbb{Z}$. On the 394 tested elements (pcgs of layers 2–4) $D_n = \gamma_n$ for $n \le 4$, consistent with Passi's theorem ($D_4 = \gamma_4$ for odd $p$).

## 5. Comparison

The reference is the answers of `dimsub.py` (kept with it, in a separate repository). Compared level by level: $|D_n|$, $|\gamma_n|$, `equal`, the set of extra elements, `index_logp`, and `certified` where it was computed in both.

**Variant B from permutations (`py/magnus.py`) and A in C (`c/dimsub_a`)**, all reference sets:

| set | groups | levels B | levels A | differences B / A |
|---|---|---|---|---|
| 8, 16, 32, 64, 81, 128, 243, 625, full mode | 2 762 | 14 961 | 16 575 | 0 / 0 |
| 729, all | 504 | 1 946 | 1 946 | 0 / 0 |
| 256, class ≥ 3 | 24 328 | 100 691 | 100 720 | 0 / 0 |
| 512, survivors of the sieve | 2 792 | 11 168 | 11 168 | 0 / 0 |
| 2187, sample | 49 | 245 | 245 | 0 / 0 |
| **total** | **30 435** | **129 011** | **130 654** | **0 / 0** |

$|D_n|$, $|\gamma_n|$, the extra elements, the certificates and the indices $\log_p[\Delta : \Delta^n + p^E\Delta]$ agree. In full mode, B on 1 425 groups (almost all elementary abelian or close to it, $M > 500$) was stopped by the rank ceiling and compared on the prefix of levels. In fast mode this happened for 29 groups of order 256 with $d = 6$. A was compared in full.

**Variant B from a presentation (`gap/export_pres.g` → `py/magnus_pres.py`)**, all elements of each group tested: orders 64, 128, 243, 729, 512 (survivors), 2187 (sample): 6 007 groups, 17 140 levels, 0 differences, indices included. Separately: 20 groups of order 64 with the exact HNF over $\mathbb{Z}$ (`--exact`) and 5 groups of order 2187 with $n_{\max} = 4 <$ cls + 1 (a check of the weight cut-off): 0 differences.

**Cross-check with an independent implementation of the same method** (unpublished; dictionaries of monomials, explicit enumeration of the pairs $u, v$, the relations of $G$ plus all commutators of weight 4).

- Their code on our groups (`scripts/xcheck_independent.py`; relations and words from our export): the same 6 007 groups and 17 140 levels, 0 differences, indices included.
- Our code on their 14 Rips-type groups $Q = G/\gamma_4(G)$ of orders $2^{40}$–$2^{107}$ (`gap/xcheck_fam.g`). The presentation is built afresh from the pc-group $Q$, not from the presentation of $G$, so the two implementations share only the method and the Howell algorithm.
  - $D_5 \cap \Omega_1(\gamma_3 \cap M_5) = 1$ in all 14 groups, as in theirs.
  - $D_4 \cap \Omega_1(\gamma_3 \cap M_4)$ has rank 1 exactly for $2^{84}$, $2^{90}$, $2^{107}$ and is trivial for the rest, as in theirs.
  - $\alpha = [e_0, e_1]^{32}[e_0, e_2]^{64}[e_1, e_2]^{128} \in D_4$ for $2^{84}$ and $2^{90}$; for $2^{107}$ and two other groups with $\alpha \ne 1$, $\alpha \notin D_4$; for the rest $\alpha = 1$. All as in theirs.

– **The exactness certificate** ($p^{E-1}A_n^+ \subseteq J_n + p^E A_n^+$) passes in all 42 cases. Therefore the memberships modulo $2^{30}$ are exact here, and $D_4(Q) \neq 1$ **for** $2^{84}$**,** $2^{90}$ **and** $2^{107}$ **is proved by computation**, not merely a "candidate modulo $2^{30}$".

**Self-contained check in this repository** (`scripts/example.sh`): GAP exports the SmallGroups of orders 64, 81 and 243 itself, and variant A, variant B from permutations and variant B from a presentation are compared with each other. It needs no reference data: 349 groups, 0 differences.

## 6. Recommendations

### 6.1. What to run

- **The original screening of the 3-groups of order** $3^7$ **is no longer needed**: all 3-groups of order $\leq 3^7$ of class 4 are metabelian, and for them $D_5 = \gamma_5$ by Gupta–Tahara 1985. For order $3^7$ it has been done anyway: B went through all 1202 candidates in 157 s on three threads, with $D_n = \gamma_n$ for all $n$ everywhere. Variant A in C independently gave the same (4 905 s on three threads): 1 202 groups, 6 010 levels, 0 differences from B. The point of the kernel now is large groups and 3-groups of order $3^8$ and above.
- **Orders** $3^8$**,** $2^{10}$ **and above.** Do not export permutations; instead export from GAP/ANUPQ the pc-presentation and only the generators of $K$ (`export_pres.g + magnus_pres.py --kernel`). The cost per group is building $J_5$ ($M = 30..780$ for $d = 2..5$, fractions of a second to seconds) plus the export in GAP. The bottleneck is generation and the sieve in GAP, not the lattice. For $d \geq 6$ and $n = 5$ ($M \geq 1554$) A in C over the whole group is better while $|G| \leq 2^{11}$.
- **The Rips group (**$2^{38}$**).** B from a presentation fits: the group has class 3, $n = 4$ is needed, and $M = \sum_{k<4} d^k \leq 399$ even for $d = 7$. This is the same as what was already computed for $3^{494}$ and for the Rips-type family $2^{40}$–$2^{107}$ (seconds per group). What is missing is a presentation of the Rips group itself: it was not found in open sources; Mikhailov–Passi, LNM 1952, is needed. Searching for the smallest quotients means adding one relation and recomputing $J_4$, also seconds.

### 6.2. What to do with the kernel next

1. Rewriting B in C/FLINT makes sense only together with the switch to standard monomials (§3.6): at the current sizes ($M \leq 780$) Python is not the bottleneck.
2. A in C with one group per thread is the fallback for $M \geq |G|$.
3. Exactness: every "$\in$" found needs a certificate (`certified`) or the HNF over $\mathbb{Z}$ (`--exact`).

### 6.3. An estimate for the plateau question of [IMP26, §3.3]: is $w \in D_7$?

$d = 6$, $n = 7$: $M = 6 + 6^2 + \cdots + 6^6 = 55\,986$. The heavy computation was not run.

**Answered in §6.5 (17.10.2026):** $w \notin D_7(G)$, certified; the dense computation on a GPU took 8 minutes.

- **Dense (as now).** One Howell form $M \times M$ in `uint32` is 12.5 GB, twice that with the row pool. The operation count is $\sim M^3/3 \approx 6 \cdot 10^{13}$ per pass. On one core (~$10^9$ operations/s in our C) that is 15–20 hours. But $J_7$ is generated not by $M$ but by $\sim \sum_r \sum_{t \leq 6-\mathrm{ord}\, r} (t+1)6^t$ rows: about $5 \cdot 10^4$ per relation with a linear part, that is, of the order of $10^6$ rows, and the closure needs several such passes. In total $10^{14}$–$10^{15}$ operations: days on the test machine, and at the limit of its memory (61 GB, shared with CI). **Not recommended.**
- **Standard monomials (§3.6).** The number of columns would be of the order of $\log_2 |\Delta/\Delta^7|$ plus the monomials with a non-invertible pivot, that is, probably thousands rather than $5.6 \cdot 10^4$. That is minutes. **Measured in §6.4: not so, at least 23 354 columns.** It needs an implementation (the nilpotent quotient of an associative algebra over $\mathbb{Z}/2^E$): a few days of work. The size can be estimated cheaply in advance: compute $\log_2 |\Delta/\Delta^n|$ for $n = 2..5$ with the current B ($M \leq 1554$) and extrapolate.
- **Without** $\Gamma$**, through** $G$**.** If $G = M/K$ is a finite 2-group, then its pc-presentation with the weight cut-off $< 7$ reduces the number of relations but not $M$. So it does not save the dense variant.

### 6.4. Measured: the size of a standard-monomial basis for the plateau group

The estimate in §6.3 ("thousands of columns") does not hold. Measured, not guessed:

- **The group.** `gap/plateau.g` builds $M$ and $\varphi$ as in Listing 1 of [IMP26] and writes $G = M/K$ as a presentation on $x_1..x_6$: the nine relators (3.12), $[[x_i, x_j], [x_k, x_l]]$, $c\, a^{-e_c}$ for the left-normed $c$ of weight 4 (this is $K$) and the commutators of

weight 5; 3 895 relators in all. A second run of nq on this presentation confirms it: class 4, $G'' = 1$, $\gamma_4(G) = \langle a \rangle \cong C_{2^{14}}$, $w = a^{2^{13}} \neq 1$. The powers $A^{-e_c}$ of the long word $A$ of $a$ are written as products of powers (`[[c,1],[A,-e]]`, read by `magnus_pres.py`), otherwise the input is 340 MB.

- $G$ **is infinite.** $G_{ab} = \mathbb{Z}^2 \oplus \mathbb{Z}/8 \oplus \mathbb{Z}/64 \oplus \mathbb{Z}/1024 \oplus \mathbb{Z}/16384$ ($x_4$, $x_6$ have no power relations); the torsion-free ranks of $\gamma_k/\gamma_{k+1}$ are $2, 1, 2, 0$. So $\Delta/\Delta^n$ is not finite and $\log_2 |\Delta/\Delta^n|$ is not defined, and the certificate of §3.2(d) does not apply; a replacement is given below.
- **The right measure** of the number of columns is the minimal number of generators of $A_n^+/J_n$, that is $\dim_{\mathbb{F}_2} A_n^+/(J_n + 2A_n^+) = \dim_{\mathbb{F}_2} \Delta_{\mathbb{F}_2}/\Delta_{\mathbb{F}_2}^n$. Any monomial basis of $\mathbb{Z}[G]/\Delta^n$, standard monomials included, has at least that many monomials. `magnus_pres.py` now prints it as `generators` ($M_n$ minus the rank of the Howell rows modulo $p$; it does not depend on $E$).
- **Jennings.** By Jennings' theorem this dimension is the sum of the coefficients at $t^1..t^{n-1}$ of $\prod_k (1 + t^k)^{d_k}$, $d_k = \dim M_k/M_{k+1}$ (`py/jennings.py`). GAP gives $G/P_7(G)$ of order $2^{347}$ ($P_7 \subseteq M_7$, so the $d_k$ for $k \leq 6$ are those of $G$) with $d_k = 6, 21, 68, 22, 0, 68$; $d_5 = 0$ because $\gamma_5(G) = 1$.

| $n$ | $M_n$ (dense) | `generators`, variant B | Jennings | time of B |
|---|---|---|---|---|
| 2 | 6 | 6 | 6 | |
| 3 | 42 | 42 | 42 | |
| 4 | 258 | 256 | 256 | 5 s |
| 5 | 1 554 | 1 226 | 1 226 | 31 s |
| 6 | 9 330 | 5 492 | 5 492 | 820 s, 13 GB |
| 7 | 55 986 | — | **23 354** | — |

Variant B and the Jennings count agree wherever B was run. The field `generators` is also checked against Jennings on all SmallGroups of orders 64, 81 and 243 ($n \leq 5$; `gap/export_jennings.g`, `scripts/compare_jennings.py`): 349 groups, 1 396 levels, 0 differences. Times are for `magnus_pres.py` (numpy, one core) on the test machine. On the side, B reproduces $w \in D_5, D_6$ and $a \notin D_5$ [IMP26, (3.15)]; for the infinite $G$ this needs the certificate below.

**Exactness for the infinite** $G$. $A_n^+/J_n \cong \mathbb{Z}^r \oplus T$, $T$ a finite 2-group with invariants $2^{v_i}$, so `index_logp` at modulus $2^E$ is $\sum_i \min(v_i, E) + Er$. The rank $r$ is known exactly: by Quillen's theorem $\operatorname{gr} \mathbb{Q}[G] \cong U(\operatorname{gr} G \otimes \mathbb{Q})$, so $r_n$ is the sum of the coefficients at $t^1..t^{n-1}$ of $\prod_i (1 - t^i)^{-h_i}$ with the torsion-free ranks $h = (2, 1, 2, 0)$: $r_n = 2, 6, 14, 27, 47, 78$ for $n = 2..7$; for $n \leq 6$ it equals the number of columns without a pivot in B. If $\operatorname{index}(E_2) - \operatorname{index}(E_1) = r(E_2 - E_1)$, then every $v_i \leq E_1$, and for an element of finite order (its image is torsion) membership modulo $2^{E_2}$, $E_2 > E_1$, is exact. Measured with `--E`:

| $n$ | $r_n$ | index, $E = 20$ | $E = 25$ | $E = 30$ | $\log_2 \lvert T \rvert$ |
|---|---|---|---|---|---|
| 2 | 2 | 73 | 83 | 93 | 33 |
| 3 | 6 | 380 | 410 | 440 | 260 |
| 4 | 14 | 1 815 | 1 885 | 1 955 | 1 535 |
| 5 | 27 | 7 286 | 7 421 | 7 556 | 6 746 |
| 6 | 47 | 28 326 | — | 28 796 | 27 386 |

The differences are exactly $5r_n$ (at $n = 6$, $10r_6 = 470$), so $T$ has exponent $\leq 2^{20}$ and the memberships modulo $2^{30}$ above are proved: $w \in D_6(G)$ and $a \notin D_5(G)$, as in [IMP26, (3.15)]. The same certificate will be needed for $n = 7$ in the dense computation.

**[computer, certified]** (D5-21, 01.11.2026). These answers are now checked by a separate program, `cert/check_in.c`, which shares no code with the computers: it rebuilds $J_n + 2^E A_n^+$ level by level (Lemma N1) by its own sparse Howell closure, checks that the result is closed under both shifts and has the Howell property, and checks the exactness above directly: the columns without a pivot are exactly $r_n$, and $2^{E-1} e_c$ lies in the span plus those columns for every $c$. Claims `results/cert/G_n{4,5,6}_E30.claim`, `G_n6_E32.claim`: $a \in D_4$, $w \in D_4, D_5, D_6$, $a \notin D_5, D_6$, and the indices of the table (1 955, 7 556, 28 796; 28 890 at $E = 32$), each in at most a second on one core (the dense B: 820 s and 13 GB at $n = 6$). Trusted: the presentation; that the rank of $A_n^+/J_n$ is at least $r_n$ (Quillen's theorem with $h = (2, 1, 2, 0)$ from nq; a smaller rank could not pass the check); and that $a$ and $w$ have 2-power order (nq: $\gamma_4(G) = \langle a \rangle \cong C_{2^{14}}$, $w = a^{2^{13}}$) — torsion of odd order would be invisible modulo $2^E$ (`cert/README.md`).

For $n = 7$ any monomial basis has at least **23 354** columns, 42 % of $M$. A standard-monomial implementation would save at most a factor of about 2.4 in columns, about 14 in the cube and 6 in memory; it is a few days of work (the nilpotent

quotient over $\mathbb{Z}/2^E$) and does not turn days into minutes. The bound is attained only if every generator can be taken to be a monomial; how many more columns the torsion costs (orders up to $2^{14}$ already in $\Delta/\Delta^2$) was not measured.

**Conclusion.** For $d = 6$, $n = 7$ the CPU route through standard monomials does not give the expected "thousands of columns"; the dense computation (on a GPU, or a CPU with $\geq 64$ GB and days) remains. For the infinite $G$ an answer "$\in$" modulo $2^E$ is exact under the certificate above; "$\notin$" is exact as before.

### 6.5. A GPU kernel for the dense Howell form modulo $2^{32}$ (prototype)

`gpu/howell_gpu.py`; for a free GPU, `gpu/howell_colab.ipynb` (Colab or Kaggle, T4 16 GB). For $p = 2$, $E \leq 32$ the arithmetic is uint32 overflow (modulo $2^E$, $E < 32$: the same and a mask).

- **Algorithm.** The Howell form of §2, blocked by panels of $b = 256$ columns. On a panel, only the rows that are nonzero there take part; the column steps run as in `c/hw.h` (pivot of least 2-adic valuation, normalised to $2^v$, exact division, the row $2^{E-v}\cdot$pivot into the pool), and every step is recorded as coefficients $G$ on the pivot rows. Then the pivot rows on the trailing columns are $(I - SG_{\mathrm{piv}})^{-1}SO$ (a unit lower triangular inverse by doubling), and every other row gets $O + G \cdot P$: one product (rows $\times b$)$\cdot$($b\times$ rest), the bulk of the work.
- **The product modulo $2^{32}$, exactly.** Three backends, each checked at start against an exact integer product: `u32` (integer matmul with wrap-around), `f32` (bytes in float32: for the shift $s$, $\sum_{i+j=s} a_i b_j$ over 64-column chunks is $< 2^{24}$, exact; 10 byte products, $i + j \leq 3$) and `i8` (signed bytes on INT8 tensor cores through `torch._int_mm`; the int32 accumulator never wraps). The last is the scheme of the Ozaki-II analogy; on the CPU only the first two run.
- **Comparison.** Two Howell forms of one span are compared after reduction to the reduced Howell form, which is unique: byte for byte, by SHA-256. Random matrices (every valuation, rank defects, $E$ from 1 to 30, panel widths 1–256; 1 500 cases): no differences from `c/hw.h`.
- **The plateau group** (input: `gpu/data/plateau_G.jsonl.gz`, written by `gap/plateau.g`). The closure of `magnus_pres.J_howell` runs on top of the blocked form, with the shifts fed two at a time and the stop rule "the index did not change" (span $H \subseteq$ span $H'$ with equal index is equality). It runs modulo $2^{32}$; the canonical forms of its images modulo $2^{30}$ and $2^{20}$ coincide with those of `magnus_pres.py` with `c/hw.h` at $E = 30, 20$, and modulo $2^{32}$ with `magnus_pres.py` with `py/howell.py` (`gpu/reference.json`), for $n = 4, 5, 6$. The numbers of §6.4 are reproduced (`generators` 256, 1 226, 5 492; $a \notin D_5, D_6$, $w \in D_4, D_5, D_6$), the free columns are Quillen's $r_n$ (14, 27, 47), and index($2^{32}$) − index($2^{20}$) $= 12r_n$: the certificate holds.
- **Measured on the CPU** (the numpy path of the same code, on the machine of §6.4): $n = 6$ closure 305 s, of which 269 s are the trailing products ($8.4 \cdot 10^{11}$ multiply-adds modulo $2^{32}$) and 275 062 column steps; `magnus_pres.py` took 820 s. From $n = 5$ to $n = 6$ the products grow as $M^{2.55}$.
- **On a GPU** (Colab, A100-SXM4 40 GB, 16.10.2026, main 93a6259): `u32` exact, 11.2 T mod-$2^{32}$ multiply-adds/s on $8192 \times 256 \times 8192$; `f32` exact, 0.67 T/s; `i8` unavailable (`CUBLAS_STATUS_NOT_SUPPORTED`). The closure at $n = 4, 5, 6$: 2.3 / 17.6 / 127.9 s, all checks pass, the canonical forms equal the CPU ones. At $n = 6$ the products take 3.2 s of 128: the rest was the column steps, 0.88 ms each (a host synchronisation and about twenty small kernels per column).
- **For** $n = 7$ ($M = 55\,986$, $r_7 = 78$, so rank 55 908 is expected):
  - *Relator rows.* `Words` builds $\mu(r) - 1$ by shifts: $v \cdot \mu(x_g^e) = \sum_k C(e,k)\, v\, a_g^k$, and $v \cdot a_g$ moves the block of degree $k$ to degree $k + 1$, one indexed write; products of powers of words still go through `Alg.mul`. On a process pool: $n = 6$ in 1.8 s instead of 54 s, the same rows byte for byte; $n = 7$ in 11 s on 8 cores, 3 894 rows, 0.8 GiB.
  - *Memory.* The form is kept by blocks (the rows of a panel only from its first column on: half of the dense $k \times M$, 5.8 GiB at $n = 7$). `howell()` takes the rows as pieces $(c_0, R)$ in increasing $c_0$ and puts a piece into the pool only at the panel containing $c_0$; before that its rows take part in no step, so the result is the same (checked: pieces against the whole matrix, the canonical forms coincide). The closure feeds the previous form block by block and frees each block as it goes; the shifts are taken only of the rows that are nonzero below the top degree (about 9 300 rows at $n = 7$, 2.1 GB per shift matrix). The peak of the closure (two shift matrices per call, measured on the CPU with `tracemalloc`): 16.8 / 2.24 / 1.88 times the dense $M^2$ uint32 matrix at $n = 4, 5, 6$, falling with $M$; at $n = 7$ that is at most $1.88 \cdot 11.7 \approx 22$ GiB [extrapolation].
  - *The first A100 run at $n = 7$ (16.10.2026, main 53658b5) ran out of memory*: rounds 0–3 in 3 / 26 / 57 / 85 s (rank 3 962 → 20 975 → 33 142 → 38 258), the pool peak 3.7 → 12.3 → 25.0 → 38.8 GiB, and round 4 failed to allocate 3.4 GB. The CPU measurement missed two things that only show at $n = 7$: the shifts were fed whole from column 0, so all their rows entered the pool at the first panel at full width; and the trailing update went by tiles of 16 384 rows, i.e. 3.7 GB per temporary at $M = 55\,986$, with four of them (the gathered rows, the product, the sum, the mask), plus three pool-sized copies when the pool was rebuilt. Now: a shift is a stream of pieces of 2 048 rows of $B$, each built when `howell()` reaches its leading column (the shifts map the columns monotonically, so the pieces come in increasing $c_0$; the streams are merged by $c_0$) — the shifted rows lead

mostly in the top degree, so they join late and narrower; the tiles are of $2^{29}$ bytes, with two temporaries; the pool is rebuilt into one new buffer; the CuPy cache is returned to the driver when it holds more than 4 GiB unused; step 8 takes one shift per call (`--group 1`). The same results on the CPU ($n = 4, 5, 6$, all hashes), the closure at $n = 6$ in 203 s instead of 368 s, the peak 1.64 instead of 1.88 dense matrices.

– **The result at** $n = 7$ (Modal, A100-SXM4 80 GB, 16.10.2026, main 9114cb9, `gpu/modal_run.py`; column steps `cuda`, one shift per call). The stages `selftest`, `random --big` (all three step modes) and $n = 4, 5, 6$ against `gpu/reference.json` pass first. The closure: rounds 0–6 in 2 / 31 / 52 / 69 / 100 / 117 / 118 s (rank 3 962 → 20 975 → 33 142 → 38 258 → 51 072 → 55 908 → 55 908; index 1 666 980 → … → 108 940 → 108 940), 488 s in all, 73 Howell calls, $5.1 \cdot 10^{14}$ multiply-adds modulo $2^{32}$ in 227 s, 3.75 M column steps; the pool peak 16.4 GiB (the round-3 peak 10.2 GiB against 38.8 GiB before); the form 5.86 GiB. Summary, canonical forms and the images modulo $2^{30}, 2^{20}$: 138 s.

| $E$ | index | rank | canonical form, sha256 |
|---|---|---|---|
| 32 | 108 940 | 55 908 | `e060b5e742633ba9…` |
| 30 | 108 784 | 55 908 | `464917aa22bfe1dd…` |
| 20 | 108 004 | 55 908 | `905de7e5675eb7b7…` |

Free columns $78 = r_7$; index$(2^{32})$ − index$(2^{30}) = 156 = 2r_7$ and index$(2^{32})$ − index$(2^{20}) = 936 = 12r_7$, so the torsion has exponent $\le 2^{20}$ and the memberships are exact: $a \notin D_7(G)$, $w \notin D_7(G)$; with §6.4, $w \in D_6(G) \setminus D_7(G)$. The rank modulo 2 gives 23 354 generators, exactly the lower bound of §6.4 for a monomial basis. [no CPU reference at $n = 7$: the certificate and the $r_7$ check are the only checks of the result.] The form modulo $2^{32}$ is kept in the Modal volume `cflint-runs`, `20261016-195219_9114cb9/J7_E32.npz`. **[computer, certified]** (D5-21, 30.10.2026). Both non-memberships now have certificates that a separate program checks: `results/cert/G_n7_{a,w}.cert.gz`, a functional $\lambda : A_7^+ \to \mathbb{Z}/2^{32}$ with $\lambda(J_7) = 0$ and $\lambda(\mu(w) - 1) \neq 0$. `cert/check_notin.c` shares no code with the computers. It builds $\mu(r) - 1$ from the presentation itself, and checks $\lambda(u(\mu(r) - 1)v) = 0$ for all monomials $u, v$. On one core this takes 6.5 s per certificate and 52 MB. Non-membership needs no exactness condition on $E$. What remains trusted is the presentation written by `gap/plateau.g` (as a presentation of $G$ modulo $\gamma_7$: it omits relators in $\gamma_7(F)$), the theorem of §3.1, and the C compiler (`cert/README.md`). The index and both answers are also reproduced by `cert/check_in.c` (01.11.2026; `results/cert/G_n7_E32.claim`): index 108 940 and 55 908 rows at $E = 32$, 78 free columns $= r_7$, in 3 s on one core.

– *Column steps.* Three implementations of the same arithmetic in the same order: `host` (as before, one synchronisation per column), `dev` (device scalars, no synchronisation; the unit inverse by Newton's iteration), `cuda` (three kernels per column: a reduction for the pivot (least valuation, then the first row), the pivot row with the column of factors, the rank-one update). On the CPU `host` and `dev` coincide with `c/hw.h` on every random matrix and with `gpu/reference.json` at $n = 4, 5$ (`dev`) and $n = 4, 5, 6$ (`host`). `cuda` is checked in the notebook (step 3) before it is used.

– *Canonical form and membership* by blocks, on the device. The form can be saved (`--save`).

– The notebook's step 8 runs $n = 7$ on a GPU with at least 38 GB. There is no CPU reference at $n = 7$: the checks are the free columns against $r_7$ and the certificate index$(2^{32})$ − index$(2^{30}) = 2r_7$, index$(2^{32})$ − index$(2^{20}) = 12r_7$.

### 6.6. D5-7: the search for a metabelian group of class 4 with $D_7 \neq 1$ (first pass)

**Reduction used** [on paper]. Let $M = \Gamma/\gamma_5(\Gamma)\Gamma''$ and $K \subseteq \gamma_4(M)$. If $1 \neq v \in D_7(M/K) \cap \gamma_4$, take $K' \supseteq K$ maximal with $v \notin K'$: then $v \in D_7(M/K')$ by functoriality and $\gamma_4(M)/K'$ is a finite cyclic 2-group (finitely generated abelian with a unique minimal subgroup). So it suffices to test the *cocyclic* quotients $G_\varphi = M/\ker\varphi$, $\varphi : \gamma_4(M) \twoheadrightarrow C_{2^m}$, and in each the single element $A^{2^{m-1}}$ of order 2 of $\gamma_4(G_\varphi) = \langle A \rangle$. A negative answer modulo $2^{32}$ ("$\notin$") is exact without any certificate ($J_7 \subseteq J_7 + 2^{32}A_7$); "$\in$" needs the certificate of §6.4.

**Pipeline.**

- `gap/docker/Dockerfile`, `scripts/gap.sh`: GAP 4 with nq from Debian in Docker (`scripts/gap.sh gap/<file>.g VAR=value ...`). Reproduces `gpu/data/plateau_G.jsonl.gz` byte for byte (as JSON).
- `gap/c4search.g`: from a spec (generators, relators, optionally a test element $w$ or explicit values of $\varphi$ on the left-normed weight-4 commutators) builds $M$ with nq, enumerates cocyclic $\varphi$ (with $\varphi(w) = 2^{m-1}$ if $w$ is given), writes $G_\varphi$ as in `gap/plateau.g` (relators, $[[x_i, x_j], [x_k, x_l]]$, $cA^{-e_c}$, $A^{2^m}$, weight-5 commutators), the torsion-free ranks $h$ for the

Quillen certificate, and flags which tested elements are nontrivial in $G_\varphi$ (lattice membership in the snf coordinates of $\gamma_4(M)$).
- Families: `gap/c4_plateau_quot.g` (Q, R, S below), `gap/c4_lie1.g` (L), `gap/c4_rand.g` (T).
- `gpu/howell_gpu.py batch --input F --n 5 6 7 --gate`: per group and $n$, membership of the tests, the certificate with the record's own $h$; `--gate` skips $n$ unless a *nontrivial* test lies in $D_{n-1}$. `gpu/modal_batch.py`: the same on a Modal A10G (24 GB is enough for $d = 6$, $n = 7$: peak 16 GiB, 8–9 min per group, as fast as the A100 for this code).
- Validation: the record of $G$ rebuilt by `c4search.g` (family R, `R0`) gives indices 1983, 7610 at $n = 4, 5$ (CPU) and 28 890, 108 940 / 108 784 / 108 004 at $n = 6, 7$ (A10G): equal to `gpu/reference.json` and to §6.5; $w \notin D_7$ again.

**Results** (raw lines in `results/c4/`). All "$\notin$" are exact; all certificates hold unless said.

| family | what |
|---|---|
| R0 | the plateau group $G$ itself, rebuilt (validation) |
| P6 | other cocyclic $\varphi$ of $M(\Gamma)$ with $\varphi(w) = 2^{m-1}$, $m = 14, 15, 16$ ($d = 6$) |
| Q | $\Gamma$ with $x_5$ killed or identified ($x_5 = 1, x_4, x_6, x_1, x_2, x_3, x_1x_2, \ldots$; also $x_6$ or $x_4$ identified; $d = 4, 5$) |
| R | quotients $G/N$ of the plateau group with the same $\varphi_0$ (or another $\varphi$ when $\varphi_0$ kills $w$): $N$ from $x_4^{2^k}$, $x_6^{2^k}$, powers of $x_i$, $d_i$, $[x_i, x_j]$, $t$ |
| R2 | $G/\langle [x_i, x_j] \rangle$ for the 11 pairs outside $\{12, 13, 23, 46\}$, and $[x_4, x_6, x_4] = [x_4, x_6, x_6] = 1$ (with $x_4^{2^{15}} = x_6^{2^{15}} = 1$) |
| S | (3.12) with every exponent $\times 2$ or $\times \frac{1}{2}$ w.p. 0.3 (16 random), $\varphi$ random with $m = 14$ |
| L | group version of $L_1$ [IMP26, Ex. 3.2], $d = 5$, exponents $\times 1, 2, 4$, (iv) with $2^6..2^8$ or none, all cocyclic $\varphi$ |
| T | random (3.12)-shaped presentations, $d = 4$ ($x_1^{2^a} \prod d_j^p$, $d_j^{2^b}$, $\prod [d_i, x_i]^q$), 40 specs $\times$ all $m \le 12$ |

So in none of the groups tested is $D_7 \cap \gamma_4 \neq 1$; the only groups with $D_6 \cap \gamma_4 \neq 1$ are the plateau group and those of its quotients where $w$ survives. Observations [computer]: the plateau is rigid — every one-exponent perturbation of (3.12) and every identification of $x_5$ kills $w$ in $M$; the naive group version of $L_1$ loses one degree ($D_5 \neq 1 = D_6 \cap \gamma_4$), i.e. the binomial terms of $x_i^{e_i}$ work *against* the plateau there; random $d = 4$ presentations of the same shape never reach $D_5 \cap \gamma_4 \neq 1$.

### 6.7. Class 5 (D5-7): shifting the plateau of [IMP26, §3.3]

The question: is there a metabelian group of class 5 with $D_8 \neq 1$ (the group analogue of the ring $L_2$ of [IMP26, Theorem A]), and then is $D_9 \neq 1$? A dense test of $D_9$ needs $d \le 4$ generators ($M = 87\,380$; $d = 5$ gives 488 280).

**Tools.** `gap/plateau5_lib.g` builds, for a presentation $\Gamma$ on $x_1..x_d$, the group $M = \Gamma/\gamma_{c+1}\Gamma''$ (nq), a projection $\varphi$ of $\gamma_c(M)$ onto one cyclic factor of its Smith basis on which $w \neq 1$, and $G = M/\ker\varphi$ written as a presentation (the relators, $[U, V]$ for left-normed $U, V$ of weights $\ge 2$ with total weight $\le c$, $C\, c_0^{-e_C}$ for the left-normed $C$ of weight $c$, and the commutators of weight $c + 1$), in the JSON of `gpu/howell_gpu.py`. The structure of $G$ (class, $G'' = 1$, $\gamma_k/\gamma_{k+1}$, torsion-free ranks for the Quillen certificate, $|w|$) is read off the pcp quotient $M/\ker\varphi$; the second nq run on the written presentation (as in `gap/plateau.g`) is optional, since nq overflows (`CombiCollect`) on these exponents. `gap/plateau5_cands.g` holds the families, `scripts/plateau5_scan.py` runs GAP and the closure, `gpu/modal_c5.py` runs both on Modal (GAP 4.14 + nq on a CPU container, the closure on an A100 80 GB), `scripts/gap5.sh` runs GAP in Docker locally. `howell_gpu.py plateau` now takes any input and any number of tests (`test_names`); the CPU reference is compared only for the default input. `random` and `plateau --n 4 5 6` on the CPU still pass against `gpu/reference.json`.

**Check on [IMP26, §3.3].** The library with $\varphi$ chosen by $a$ (Listing 1) gives $w \in D_6$, $a \notin D_6$ (certified), as in §6.4–6.5.

**Which relators of (3.12) matter** ($n = 6$, all certified). Dropping the fourth relator $x_5^{1024}d_2^{16}$ loses $w \in D_6$; dropping any one of the other five ($d_5^{2048}$, $d_1^{16}$, $d_2^{128}$ and the two top relators) keeps it, and so does dropping all five at once. Without $x_5$ altogether $w \notin D_6$; without $d_5^{16}$ in the second relator $w = 1$. The free generator $x_6$ can be replaced by $x_3$ ($t = [x_4, x_3]$): five generators, $w \in D_6$. So the class-4 plateau survives on five generators with four relators:

$$x_1^8 d_2 d_3^4, \quad x_2^{64} d_1^{-1} d_3^{-16} d_5^{16}, \quad x_3^{16384} d_1^{-4} d_2^{16}, \quad x_5^{1024} d_2^{16}, \qquad d_i = [x_4, x_3, x_i].$$

**Four generators, class 4: every attempt has $w \notin D_6$ or $w = 1$.** The Rips-type group of [BM23, §1.2] with $z$ replaced by $t = [z, e_2]$, and its scalings: $w \in D_5 \setminus D_6$ (also for its greedy maximal quotient); (3.12) without $x_5$, $t = [x_4, x_3]$: $w \in D_5 \setminus D_6$; $x_5$ merged into the generator of $t$, or $x_5 := W(x_1..x_4)$ for eight words $W$, or $t$ built from torsion generators only (ten choices): $w = 1$ in $M$.

**Class 5.** $t = [x_4, x_3, x_3]$, $d_i = [t, x_i] \in \gamma_4$, the same relators, $c = 5$, $\gamma_5(G) \cong C_{2^{14}}$:

| group | $d$ | $D_6$ | $D_7$ | $D_8$ |
|---|---|---|---|---|
| [BM23]-type, $t = [z, e_2, e_2]$ (three scalings) | 4 | $w \in$ | $w \notin$ | |
| (3.12) without $x_5$ | 4 | $w \in$ | $w \notin$ | |
| (3.12) shifted, five generators (`plateau5_c5i5`) | 5 | $w \in$ | $w \in$ | $w \notin$ |
| the same with four relators; with $L_{13}, L_{23}$ doubled; with $E_5 = 512$ | 5 | $w \in$ | $w \in$ | — |

All entries certified (free columns $= r_n$, index differences $2r_n$, $12r_n$). For `plateau5_c5i5` at $n = 8$: $M = 97\,655$, closure in 7 rounds and 1 829 s on an A100 80 GB (peak 58.8 GiB), rank 97 648, free columns $7 = r_8$, index 169 278, index$(2^{32})$ − index$(2^{30}) = 14$, −index$(2^{20}) = 84$. Since $\gamma_5(G)$ is cyclic and $w$ is its element of order 2, $w \notin D_8$ gives $D_8 \cap \gamma_5 = 1$; for the four-relator version the tests $u_3^{2^j}$ give $D_6 \cap \gamma_5 = \langle w \rangle$. So this group has $D_7 \neq 1$ ($f_{\mathrm{met}}(5) \geq 7$), not $D_8 \neq 1$.

**Why the shift stops at $D_7$ (the pairing lemma).** If $x^E \in \gamma_k$, $E = 2^s \geq 4$, then $E(x-1) \in \Delta^2$, $2E(x-1) \in \Delta^3$, $4E(x-1) \in \Delta^4$ (expand $x^E - 1 = \sum_m \binom{E}{m}(x-1)^m$; for the last step $4\binom{E}{3} = \frac{(E-1)(E-2)}{3} \cdot 2E$ and $3 \mid (E-1)(E-2)$). Hence $(x-1)(\zeta^{2^{k-2}E} - 1) \in \Delta^{2k}$ for $\zeta \in \gamma_k$, $k \leq 4$: this is the factor $x_1^{16} = x_1^{2E_1}$, $x_2^{128} = x_2^{2E_2}$ in (3.14) ($k = 3$). For class 5 ($k = 4$) the analogue needs $g_i = \prod_j x_j^{L_{ij}} = \zeta_i^{4E_i}$, while (3.12) has $g_1 = \zeta_1^{16}$, $g_2 = \zeta_2^{128}$, i.e. $2E_i$. `scripts/plateau5_pairing.py` enumerates the power-of-2 scalings of $E_i$, of the rows of the exponent matrix and of $L_{ij}$ for which $g_i \in 4E_iD$ modulo $\gamma_c$ (121 644 of them); the 80 cheapest, built at class 4, all have $w = 1$ in $M$. In this family the two requirements are in tension: $w \neq 1$ needs some $L_{ij} < E_iE_j$ (since $[x_j, x_i^{E_i}]$-type factors have order dividing $E_j$), and the quadrupled pairing pushes the $L_{ij}$ up.

**Lie rings on four generators** (on paper, by the proofs of [IMP26, §3.1–3.2]). In $L_N$ the free generator $x_5$ can be replaced by $x_3$: on $x_1..x_4$ with $x_3 < x_1 < x_2 < x_4$, $t = [x_4, x_3, \dots, x_3]$ ($N$ letters $x_3$), the same relators $r_i$ and (ii)–(iv), one gets $D_{N+4} = \cdots = D_{2N+4} = \langle a \rangle \cong \mathbb{Z}/2$, $D_{2N+5} = 0$. The places that change: in the lemma on the top layer a weight-$(N+3)$ bracket $e_3[x_3, \dots]$ can now hit $u_1, u_2$, but $64\mu(u_1)$, $64\mu(u_2) \equiv 0 \bmod 256$; in the lemma on the last initial form the $c_i$-coordinate of a bracket of $r_3$ is a multiple of $e_3 = 64$, which $e_1, e_2$ divide. For $N = 2$: class 5, $D_6 = D_7 = D_8 \neq 0 = D_9$ on four generators. The group analogue of this ring is what the table above fails to produce.

**Rerun.**

```
# the class-5 group (five generators); about 9 min, Docker, image from gap/Dockerfile.plateau5
scripts/gap5.sh plateau5_cands.g 'P5NAME := "c5i5"; P5I5 := rec(cls := 5); P5OUT := "/data/c5i5.jsonl";'
# D_n on a GPU with 64 GB or more for n=8 (the committed plateau5_c5i5.jsonl.gz is that file)
python3 gpu/howell_gpu.py plateau --input gpu/data/plateau5/plateau5_c5i5.jsonl.gz --n 6 7 8 --h 1 0 0 0 0 --
# on Modal (detached; outputs in the volume math43-c5)
modal deploy gpu/modal_c5.py
python gpu/modal_c5.py cpu|gpu '<shell command>' ...
# a scan of candidates (SCAN_DEVICE=gpu; SCAN_REUSE=1 reruns only the closure)
python3 scripts/plateau5_scan.py LOG 'name|I5rec(cls:=4, ...)|5,6'
```

Modal spend for all of §6.7: about 8 USD. A greedy maximal quotient of the class-5 group keeping $w$ (`P5_Quot`, 65 bases) did not finish in 1 h 45 min of GAP and was stopped.

### 6.8. D5-7, class 4: the gap $D_7 \not\subseteq \gamma_4$, and all $K \subseteq \gamma_4(M)$ at once

Two additions to §6.6. (a) closes the case $D_7 \not\subseteq \gamma_4$ that §6.6 left open, for every group tested there. (b) replaces the enumeration of $\varphi$ by one linear test that covers every $K \subseteq \gamma_4(M)$, and says where the exact question stops being linear. Data: `results/c4/q3/` (a), `results/c4/kfree/` (b).

**(a) The class-3 quotient** **Lemma 1** [on paper]. Let $Q$ be a finitely generated nilpotent group of class $\leq 3$ and $n \in \{4, 5\}$. Then $D_n(Q) \subseteq \tau(\gamma_3(Q))$ (the torsion of $\gamma_3$), and $\theta : \gamma_3(Q) \to \Delta^3/\Delta^n$, $g \mapsto g - 1$, is a homomorphism with kernel $D_n(Q) \cap \gamma_3(Q)$. Hence, if $\tau(\gamma_3(Q))$ is a 2-group, $D_n(Q) = 1$ iff $\theta$ is injective on the elements of order 2 of $\tau(\gamma_3(Q))$. *Proof.* $D_3 = \gamma_3$ for every group (classical, see [MP09, ch. 1] †), so $D_n(Q) \subseteq \gamma_3(Q)$. By Jennings the rational dimension subgroups are the isolators: $D_n(Q) \subseteq \sqrt{\gamma_n(Q)} = \tau(Q)$ for $n \geq 4$ (Jennings 1955 †; the same fact as Quillen's theorem behind the certificate of §6.4). For $g, h \in \gamma_3$: $gh - 1 = (g-1) + (h-1) + (g-1)(h-1)$ and $(g-1)(h-1) \in \Delta^6$. A nontrivial subgroup of a finite 2-group contains an element of order 2. □

**Corollary 2** [on paper]. Let $G$ be a group with $\gamma_5(G) = 1$ and $Q = G/\gamma_4(G)$ with $D_5(Q) = 1$. Then $D_r(G) \cap \gamma_4(G) = 1$ implies $D_r(G) = 1$ for every $r \geq 5$: the image of $D_r(G)$ in $Q$ lies in $D_r(Q) \subseteq D_5(Q) = 1$.

For a record $G_\varphi = M/\ker\varphi$ of §6.6 one has $G_\varphi/\gamma_4 = M/\gamma_4(M) = \Gamma/\gamma_4(\Gamma)\Gamma''$: $Q$ depends only on the base presentation $\Gamma$, not on $\varphi$. So 80 groups $Q$ cover all 844 groups of §6.6.

**Computation** [computer]. `scripts/c4_q3.py` takes the relators of $\Gamma$ from a record (dropping the $\varphi$-dependent ones), `gap/c4_q3.g` builds $Q$ with nq ($Q'' = 1$ is asserted) and writes the exponents of the left-normed $[x_i, x_j, x_k]$ in the snf basis of $\gamma_3(Q)$; `scripts/c4_q3.py prep` solves them (modulo $2^{32}$, checked with exact integers) for every element of order 2 of the torsion, as a combination of the commutators (the row of the element is the same combination of rows: additivity, $n \le 2c$); then `howell_gpu.py batch --span` at $n = 4, 5$ computes $\log_2 |\langle\text{tests}\rangle|$ in $A_n/(J_n + 2^E A_n)$ as index$(J)$ − index$(J +$ tests$)$. (The committed lines were made by an earlier version of the script that wrote each element as a word, found in GAP; for P6 the present version gives the same: 68 of 68 at $n = 4, 5$.) Injectivity found modulo $2^{32}$ is exact without any certificate (a relation in $J$ is a relation in $J + 2^{32}A$); the certificates hold anyway.

| bases | $d$ | $\gamma_3(Q)$ | elements of order 2 | result |
|---|---|---|---|---|
| P6 = R0 (the plateau $\Gamma$) | 6 | $\mathbb{Z}^2 \oplus C_8^{30} \oplus C_{64}^{20} \oplus C_{1024}^{12} \oplus C_{16384}^{6}$ | 68 | $\theta$ injective at $n =$ |
| R (Rf15, Rf16, Rd4, Rd6) | 6 | finite 2-group (Rf15, Rf16: $Q$ itself finite) or $\mathbb{Z}\oplus$ 2-group | 68–70 | injective |
| R2 (12 bases) | 6 | $\mathbb{Z}^2\oplus$ 2-group, or finite (Rd46, Rd46f15; $Q$ finite for Rd46f15) | 62–68 | injective |
| S (16) | 6 | $\mathbb{Z}^2\oplus$ 2-group | 68 | injective |
| L (7) | 5 | $\mathbb{Z}^2 \oplus C_4^{20} \oplus C_{16}^{12} \oplus C_{64}^{6}$ (scaled for ×2, ×4) | 38 | injective |
| T (40) | 4 | finite 2-group ($Q$ infinite, $h_1 \ge 1$) | 20 | injective |

In all 80, $\tau(\gamma_3(Q))$ is a 2-group and $\theta$ is injective already at $n = 4$: $D_4(Q) = 1$ (so $D_5(Q) = 1$). Lines: `results/c4/q3/q3_*.out.jsonl` (`span_bits` = `tests` at $E = 32, 30, 20$).

**What is now proved** [computer + on paper, Lemma 1, Corollary 2, and the tests of §6.6]:

- the plateau group $G$ of [IMP26, §3.3] (R0) and the 3 groups P6: $D_7 = 1$. For $G$ this no longer uses [IMP26]'s description of $D_6(G)$;
- the 16 surviving quotients R, R2 of the plateau (Rf15, Rf16, Rd4, Rd6 and the 12 of R2): $D_7 = 1$;
- the 62 groups L: $D_6 = 1$; the 54 of them with $A^{2^{m-1}} \notin D_5$: $D_5 = 1$;
- the 16 groups S and the 746 groups T: $D_5 = 1$.

In each case "the group" is the cocyclic quotient $G_\varphi$ that was tested, not $M$ and not $\Gamma$: only a few $\varphi$ were tried per $M$. Part (b) is about $M$ and all its quotients.

**Relation to D5-2** [on paper]. An element of $D_7(G) \setminus \gamma_4(G)$ in a class-4 group gives $1 \ne x \in D_5(Q)$ for the class-3 quotient; $x$ has 2-power order, and a finitely generated nilpotent group is residually a finite $p$-group on its $p$-elements (Gruenberg †), so $x$ survives in a finite 2-group of class $\le 3$ with $D_5 \ne 1$: a negative answer to D5-2 (Kourovka 20.84). So the gap closed here is exactly an instance of D5-2; the 80 groups $Q$ (three of them, Rf15, Rf16, Rd46f15, finite 2-groups of class 3) all have $D_4 = 1$, stronger than the $D_5 = 1$ that D5-2 asks for. They are instances, not evidence of a general argument.

**(b) Which $K \subseteq \gamma_4(M)$ could give $w \in D_7(M/K)$: one Howell form for all of them** Let $M = \Gamma/\gamma_5(\Gamma)\Gamma''$ (class 4), $A_4 = \gamma_4(M)$, $A = \mathbb{Z}[M]$, $\Delta$ its augmentation ideal, $n \le 8$.

**Lemma 3** [on paper]. (i) $\psi : A_4 \to \Delta^4/\Delta^n$, $k \mapsto k - 1$, is a homomorphism ($(k-1)(k'-1) \in \Delta^8$). (ii) For $K \subseteq A_4$ and $v \in M$: $v \in D_n(M/K) \iff v - 1 \in \Delta^n + (K-1)A$, and

$$(K-1)A + \Delta^n = \Delta^n + \sum_k \psi(k)\cdot R, \qquad R = \operatorname{span}\{1\} \cup \{(x_{j_1} - 1)\cdots(x_{j_l} - 1) : 1 \le l \le n-5\},$$

$k$ over generators of $K$ ($K$ is central, $\psi(k)\Delta^{n-4} \subseteq \Delta^n$, and $\Delta/\Delta^{n-4}$ is spanned by those monomials). *Proof.* The kernel of $\mathbb{Z}[M] \to \mathbb{Z}[M/K]$ is $(K-1)A$, and $\Delta(M/K)^n$ is the image of $\Delta^n$. For $k = \prod k_j^{e_j}$, $\psi(k) \equiv \sum e_j\psi(k_j)$ by (i). □

So for a fixed $K$ the question $w \in D_n(M/K)$ is linear: add the rows $\psi(k_j)\cdot r$ to $J_n(M)$. The existence of *some* $K$ is not: with $\Lambda(k \otimes r) = \psi(k)r$, a $K$ works iff some solution $(k_r)_r \in A_4^R$ of $\sum_r \psi(k_r)r \equiv \psi(w)$ (an affine coset of $\ker\Lambda$) has $w \notin \langle k_r\rangle$. "The coordinates of a point of a coset generate a subgroup that misses $w$" is a union over the subgroups $K$ (equivalently over $\varphi \in \widehat{A_4}$ with $\varphi(w) \ne 0$), not an intersection: dually, a counterexample is a $\varphi$ that is not of the form $\varphi = \chi \circ \Lambda(\cdot \otimes 1)$ for a character $\chi$ of $A/J_n$ with $\chi\circ\Lambda \in \langle\varphi\rangle \otimes \widehat{R}$ — a rank-one condition on a tensor, which is not linear in $\chi$. This is where a single linear problem stops. What does survive is one-sided:

**Proposition 4** [on paper]. Let $U_n = J_n + \Lambda(A_4 \otimes R_+)$, $R_+$ the monomials of degree $1..n-5$, and $\bar\psi : A_4 \to A/U_n$. If $\bar\psi$ is injective, then $D_n(M/K) \cap \gamma_4(M)/K = 1$ for **every** $K \subseteq A_4$. *Proof.* If $v \in D_n(M/K) \cap A_4$, $v \notin K$, then by Lemma 3 $\psi(v) \in J_n + \psi(K) + \Lambda(K \otimes R_+) \subseteq U_n + \psi(K)$, so $\bar\psi(vk^{-1}) = 0$ for some $k \in K$, and $v = k \in K$. □

Conversely, $\ker\bar\psi$ contains $D_n(M) \cap A_4$ (take $K = 1$), and every $v \in D_n(M/K) \cap A_4$, $v \notin K$, gives $1 \neq vk^{-1} \in \ker\bar\psi$ for some $k \in K$ — the test is a filter: it can only fail where a counterexample *could* be. Two practical points:

- $\ker\bar\psi$ is torsion: $\Lambda(A_4 \otimes R_+) \subseteq \Delta^5$, so $z \in \ker\bar\psi$ gives $z \in D_5(M) \subseteq \sqrt{\gamma_5(M)} = \tau(M)$ (Jennings), so it is enough to test the elements of order 2 of $\tau(A_4)$ (all of them at once: `socle_bits` = their number); injectivity found modulo $2^{32}$ is exact (as in (a)). Since $\ker\bar\psi_{n+1} \subseteq \ker\bar\psi_n$, injectivity at $n$ gives it at every larger $n \leq 8$.
- With $D_4(M/\gamma_4) = 1$ from (a) and Corollary 2: $\bar\psi_n$ injective $\Rightarrow D_n(M/K) = 1$ for every $K \subseteq \gamma_4(M)$ ($n \leq 7$).

**Implementation.** `gap/c4_q3.g` with `Q3_C=4` writes $M$ with the exponents of the 540 (for $d = 6$) left-normed commutators of weight 4 in the snf basis of $\gamma_4(M)$; `scripts/c4_q3.py prep --kgens` solves for the snf basis $b_t$ (the generators of $A_4$) and for the elements of order 2, modulo $2^{32}$; `howell_gpu.py batch --kfree` builds $U_n$ (the closure $J_n$, then one more Howell form with the rows $\psi(b_t)\,a_{j_1}\cdots a_{j_l}$, consuming $J_n$) and reports `socle_bits` and the membership of every test in $U_n$ (`kfree` in the output lines). Rerun: `python3 scripts/c4_q3.py gapin X.g gpu/data/c4/P6.jsonl.gz`, `scripts/c4_q3_docker.sh 4 X.g OUT` (or `scripts/c4_q3_gaprun.sh` with a native GAP), `python3 gpu/howell_gpu.py batch --input OUT.jsonl --n 5 6 --kfree`; for $n = 7$, $d = 6$: `C4BATCH_APP=<name> modal deploy gpu/modal_batch.py`, then `gpu/modal_batch.py spawn --gpu a10g ... --n 7 --kfree --group 1`. (The T and L lines were made by the earlier version, which wrote each generator as a word; L rebuilt with the present version gives the same numbers.) GAP in Docker on the test machine: a run of the earlier version (preimage words) on the plateau $M$ exhausted the memory of the Docker VM and hung the daemon for about an hour; with the present version a base takes 0.4 s ($d = 5$) to 2 min ($d = 6$). The plateau $M$ and S2, S3 were built on a Modal CPU container (`gpu/modal_c5.py` with `C5_APP`, `C5_VOL`, `C5_CPU`, `C5_MEM`).

**Results** [computer].

| $M$ | $A_4 = \gamma_4(M)$ | $n = 5$ ($U_5 = J_5$) |
|---|---|---|
| T (40 bases, $d = 4$) | 2-groups of rank 41–45 | injective in all 4 |
| L1, L1b128, L1b256 | $C_{64}$ | not injective ($A$ |
| L1x2, L1x4 | $C_{64}$ | injective |
| L1free, L1x2free | $C_4 \times C_{64}$, $C_4 \times C_{128}$ | rank 1 of 2 |
| S (16 bases, $d = 6$) | $\mathbb{Z}^3\oplus$ 2-group, 205 elements of order 2 | injective in all 1 |
| **P6: the plateau** $\Gamma$ ($d = 6$) | $\mathbb{Z}^3 \oplus C_4^2 \oplus C_8^{102} \oplus C_{16} \oplus C_{32} \oplus C_{64}^{57} \oplus C_{256} \oplus C_{512} \oplus C_{1024}^{28} \oplus C_{16384}^{12}$, 205 elements of order 2 | kernel $\langle w\rangle$ |

**The plateau,** $n = 7$ [computer] (Modal A10G, main of this branch `4107648`, 890 s, GPU peak 20.2 GiB; line `results/c4/kfree/m4_P6_n7.out.jsonl`): $J_7(M)$ has index 149 336 modulo $2^{32}$, free columns $99 = r_7$ for $h = (2,1,2,3)$, $\text{index}(2^{32}) - \text{index}(2^{30}) = 198 = 2r_7$, $-\text{index}(2^{20}) = 1188 = 12r_7$ (certificate holds); $U_7$ adds 8 736 rows $\psi(b_t)a_j$, $\psi(b_t)a_ja_{j'}$ for the 208 generators $b_t$ of $\gamma_4(M)$, index 109 840; the 205 elements of order 2 of $\tau(\gamma_4(M))$ generate a subgroup of order $2^{205}$ modulo $U_7$. At $n = 5, 6$ (CPU, `m4_P6_cpu.out.jsonl`) the rank is 204 of 205 and $w \in J_n(M)$: $D_5(M) \cap \gamma_4 = D_6(M) \cap \gamma_4 = \langle w\rangle$ (certificates hold), and $w$ is the whole kernel of $\bar\psi_5, \bar\psi_6$ — as it must be, since $w \in D_6(G)$.

**Conclusion** [computer + on paper, Proposition 4, Corollary 2, (a)]. For every $K \subseteq \gamma_4(M)$, $M = \Gamma/\gamma_5(\Gamma)\Gamma''$ with $\Gamma$ the plateau presentation (3.12): $D_7(M/K) = 1$. That is, no metabelian group of class 4 of the form $\Gamma/N$ with $\gamma_5(\Gamma)\Gamma'' \subseteq N \subseteq \gamma_4(\Gamma)\gamma_5(\Gamma)\Gamma''$ has $D_7 \neq 1$ — all central modifications of the plateau group at once, including all $\varphi$ that §6.6 did not try. Likewise $D_5(M/K) = 1$ for every $K$ for the 56 bases T and S ($D_5$ also for L1x2, L1x4), and $D_6(M/K) = 1$ for the 7 bases L. The enumeration of $\varphi$ in §6.6 for these bases was therefore not a sample but complete.

**Quotients of the plateau by relators outside** $\gamma_4$ (families R, R2 of §6.6: a different $M$ each) [computer]. At $n = 6$ (CPU, `m4_R.out.jsonl`) all 11 built bases (Rk14, Rk15, Rk16, Rk24, Rk26, Rk34, Rk35, Rk36, Rk45, Rk56, Rd46) have a kernel of rank exactly 1 (182 of 183, 183 of 184, 196 of 197) — the place of $w$, as for the plateau. At $n = 7$ one base was run, **Rd46** ($[x_4,x_6,x_4] = [x_4,x_6,x_6] = 1$, the strongest combination of §6.6 that keeps $w$; $\gamma_4(M)$ finite, 197 elements of order 2): Modal A10G, 885 s, GPU peak 20.5 GiB; $J_7$ index 142 032, free columns $49 = r_7$ for $h = (2,1,0,0)$, differences $98 = 2r_7$ and $588 = 12r_7$; $\bar\psi_7$ **injective** (197 of 197). With (a) ($D_4(Q) = 1$ for Rd46): $D_7(M/K) = 1$ for every $K \subseteq \gamma_4(M)$ for this base too.

**The other 10 R2 bases at** $n = 7$ [computer]. Same procedure as for P6 and Rd46: $M$ built by `gap/c4_q3.g` (`Q3_C=4`) on a Modal CPU container (8 cores; 4 parallel GAP streams, 9.6–10.1 min per base; logs `results/c4/kfree/gap_m4_R2n7{a,b,c,d}.log`;

input `python3 scripts/c4_q3.py gapin` from `gpu/data/c4/R2a.jsonl.gz`, `R2b.jsonl.gz`), then one A10G run per base, `gpu/modal_batch.py spawn --ref 8ed65af --gpu a10g ... -- --n 7 --kfree --group 1` (app `math43-r2n7`), 804–845 s each. Lines `results/c4/kfree/m4_<base>_n7.out.jsonl`. In all ten: $h = (2,1,2,3)$, $r_7 = 99$, $\dim A_7 = 55\,986$; the certificate of $J_7$ holds with free columns $99 = r_7$, index($2^{32}$) − index($2^{30}$) = $198 = 2r_7$, index($2^{32}$) − index($2^{20}$) = $1188 = 12r_7$; no element of order 2 of $\tau(\gamma_4(M))$ lies in $J_7$ (`in_D` all false), and $\bar{\psi}_7$ is **injective**:

| base | relator added to (3.12) | gens of $\gamma_4(M)$ | elements of order 2 | rows $\psi(b_t)a$ | index $J_7$ ($2^{32}$ / $2^{30}$ / $2^{20}$) | index $U_7$ | rank o |
|---|---|---|---|---|---|---|---|
| Rk14 | $[x_1, x_4]$ | 186 | 183 | 7 812 | 137 835 / 137 637 / 136 647 | 101 781 | 183 of |
| Rk15 | $[x_1, x_5]$ | 186 | 183 | 7 812 | 137 907 / 137 709 / 136 719 | 101 786 | 183 of |
| Rk16 | $[x_1, x_6]$ | 186 | 183 | 7 812 | 137 835 / 137 637 / 136 647 | 101 781 | 183 of |
| Rk24 | $[x_2, x_4]$ | 186 | 183 | 7 812 | 132 116 / 131 918 / 130 928 | 97 590 | 183 of |
| Rk26 | $[x_2, x_6]$ | 186 | 183 | 7 812 | 132 116 / 131 918 / 130 928 | 97 590 | 183 of |
| Rk34 | $[x_3, x_4]$ | 187 | 184 | 7 854 | 127 666 / 127 468 / 126 478 | 94 254 | 184 of |
| Rk35 | $[x_3, x_5]$ | 187 | 184 | 7 854 | 128 896 / 128 698 / 127 708 | 95 148 | 184 of |
| Rk36 | $[x_3, x_6]$ | 187 | 184 | 7 854 | 127 666 / 127 468 / 126 478 | 94 254 | 184 of |
| Rk45 | $[x_4, x_5]$ | 187 | 184 | 7 854 | 128 879 / 128 681 / 127 691 | 95 147 | 184 of |
| Rk56 | $[x_5, x_6]$ | 187 | 184 | 7 854 | 128 879 / 128 681 / 127 691 | 95 147 | 184 of |

(The pairs Rk14/Rk16, Rk24/Rk26, Rk34/Rk36, Rk45/Rk56 give equal numbers — presumably the symmetry $x_4 \leftrightarrow x_6$ of (3.12); not checked.) At $n = 6$ (CPU, `m4_R.out.jsonl`) each of them had a kernel of rank 1; at $n = 7$ the kernel is gone, exactly as for the plateau. With (a) ($D_4(Q) = 1$ for all R2 bases) and Proposition 4: $D_7(M/K) = 1$ **for every** $K \subseteq \gamma_4(M)$ for each of these 10 bases. (The commit `8ed65af` no longer prints the per-round GPU memory; the runs fit in the 24 GB of the A10G.)

**The last 5 bases: R (Rd4, Rd6, Rf15, Rf16) and Rd46f15 at** $n = 7$ [computer]. Same procedure. $M$ built by `gap/c4_q3.g` (`Q3_C=4`, with `Q3_NQTOP := false`, see below) on a Modal CPU container (5 cores / 28 GiB, five GAP streams in parallel with a limit of 1900 s each, `scripts/c4_rest_gap.sh`; input `python3 scripts/c4_q3.py gapin` from `gpu/data/c4/R.jsonl.gz` and the Rd46f15 line of `R2b.jsonl.gz`, one file `m4_in_<base>.g` per base; logs `results/c4/kfree/gap_m4_<base>.log`; app `math43-c4rest-gap`), 10–15 min per base. Then two A10G calls (`gpu/modal_batch.py spawn --ref efecbc5 --gpu a10g ... -- --n 7 --kfree --group 1`, app `math43-c4rest`; Rd4+Rd6 in one call, Rf15+Rf16+Rd46f15 in the other), 820–901 s per base, GPU peak 20.2–20.9 GiB. Lines `results/c4/kfree/m4_<base>_n7.out.jsonl`, logs `m4_Rd4_Rd6_n7.out.log`, `m4_Rf15_Rf16_Rd46f15_n7.out.log`. $\dim A_7 = 55\,986$ in all five; no element of order 2 of $\tau(\gamma_4(M))$ lies in $J_7$ (`in_D` all false: $D_7(M) \cap \gamma_4 = 1$), and $\bar{\psi}_7$ is **injective** in all five:

| base | relators added to (3.12) | $M$ | $h$ | gens of $\gamma_4(M)$ | elements of order 2 | rows $\psi(b_t)a$ | index $J_7$ ($2^{32}$ / $2^3$ |
|---|---|---|---|---|---|---|---|
| Rd4 | $[x_4, x_6, x_4]$ | infinite | $(2,1,1,1)$ | 202 | 201 | 8 484 | 145 421 / 145 28 |
| Rd6 | $[x_4, x_6, x_6]$ | infinite | $(2,1,1,1)$ | 202 | 201 | 8 484 | 145 421 / 145 28 |
| Rf15 | $x_4^{2^{15}}, x_6^{2^{15}}$ | finite | 0 | 208 | 208 | 8 736 | 147 440 / 147 44 |
| Rf16 | $x_4^{2^{16}}, x_6^{2^{16}}$ | finite | 0 | 208 | 208 | 8 736 | 147 662 / 147 66 |
| Rd46f15 | $[x_4, x_6, x_4], [x_4, x_6, x_6], x_4^{2^{15}}, x_6^{2^{15}}$ | finite | 0 | 197 | 197 | 8 274 | 141 071 / 141 07 |

(The relators of Rf15, Rf16, Rd46f15 are as in §6.6; "finite": $\gamma_4(M)$ has no free part and $h = 0$, so $M$ is a finite 2-group.) The failed certificate of Rf16 is the same as in §6.6 (there for $G_\varphi$): $A/J_7$ has elements of order $> 2^{20}$, so a Howell form modulo $2^{20}$ loses information; it does not matter here, because every answer is "$\notin$"/injective, which is exact modulo $2^{32}$ without a certificate (a relation modulo $U_7$ is one modulo $U_7 + 2^{32}A$). Rd4 and Rd6 give equal numbers (again presumably $x_4 \leftrightarrow x_6$; not checked). With (a) ($D_4(Q) = 1$ for these bases) and Proposition 4: $D_7(M/K) = 1$ **for every** $K \subseteq \gamma_4(M)$ for each of the five.

*Why nq "took hours" before* [computer]. Not reproduced. `gap/c4_q3.g` now prints the wall time per stage. On the Modal container (GAP 4.14 from Debian trixie, nq from `gap-nq`), with the weight-5 commutators *not* passed to nq (`Q3_NQTOP := false`; they are redundant for `class := 4`, and they stay in the output relators): nq takes 4 s for the infinite Rd4, Rd6 and 132 s (Rf15, Rd46f15), 267 s (Rf16) for the finite ones — the large exponents $2^{15}, 2^{16}$ cost nq a factor 30–60, but minutes, not hours. The rest of GAP is the same for all: lower central series about 1 s, abelian invariants and the snf basis of $\gamma_4$ 30–43 s, exponents of the 540 commutators 8 s, and then about 570 s (90 % of GAP's CPU time) for the one remaining step, the check that the extra test word $w$ lies in $\gamma_4(M)$ and is nontrivial (`MappedWord` of a long word into the pcp group

and membership in $\gamma_4$) — that, not nq, is what makes a base take 10 min (presumably also for the R2 bases above, whose logs have no stage times). Whether the hours in Docker came from the 3240 redundant relators given to nq for a finite group, or from the memory of the Docker VM, was not tested.

So for the first pass of §6.6 the test "for all $K$" is now done at the decisive $n$ for **every base**: the plateau P6, the 12 R2 bases (Rk14, …, Rk56, Rd46, Rd46f15) and the 4 R bases (Rd4, Rd6, Rf15, Rf16) — $D_7(M/K) = 1$; the 40 T and 16 S bases — $D_5(M/K) = 1$; the 7 L bases — $D_6(M/K) = 1$. **Summary of the first pass of class 4** [computer + on paper]: for each of the 80 bases $\Gamma$ of §6.6 and every $K \subseteq \gamma_4(M)$, $M = \Gamma/\gamma_5(\Gamma)\Gamma''$, the group $M/K$ has $D_7 = 1$ (P6, R, R2: 17 bases), $D_6 = 1$ (L: 7) or $D_5 = 1$ (T, S: 56). In particular no metabelian group of class 4 obtained from these presentations by a central modification of $\gamma_4$ has $D_7 \neq 1$; finding one needs a base $\Gamma$ outside this family.

**Next.** (1) Done for all bases of the first pass (above); (2) when $\bar\psi_n$ is not injective, the kernel is the list of candidates: for $v \in \ker\bar\psi_n$ a representation $\psi(v) = \sum \psi(k_j)m_j$ modulo $J_n$ with $v \notin \langle k_j\rangle$ is a counterexample $M/\langle k_j\rangle$ — this needs the coefficients of the representation (an augmented Howell form), not implemented; (3) the same test for class 5 ($\gamma_5$ central, $\psi$ additive modulo $\Delta^{10}$, $R_+$ of degree $\le n-6$) for the group of §6.7.

**[computer, certified]** (D5-21, 01.11.2026). Every index of $J_n$ and $U_n$ and every socle count of this subsection (modulo $2^{32}$) is confirmed by `cert/check_in.c`, which shares no code with `howell_gpu.py` (260 claims in `results/cert/c4/`, `cert/README.md`): (a) for all 80 groups $Q$ at $n = 4, 5$, the elements of order 2 of $\tau(\gamma_3(Q))$ are independent modulo $J_n + 2^{32}A_n^+$ (so $\theta$ is injective and $D_4(Q) = 1$); (b) for $M$ of every base, index($J_n$), index($U_n$) and the socle count as in the tables: T, S at $n = 5$ and L at $n = 5, 6$ (incl. the non-injective L1, L1b128, L1b256 at $n = 5$ and rank 1 of 2 for L1free, L1x2free), P6 at $n = 5, 6$ (204 of 205, $w \in J_5, J_6$) and $n = 7$ (205 of 205), Rk14, …, Rk56 and Rd46 at $n = 6$ (rank one less than full), and Rk14, …, Rk56, Rd46, Rd46f15, Rd4, Rd6, Rf15, Rf16 at $n = 7$ (**injective**), each $n = 7$ base in about 15 s on one core (A10G: 800–900 s). All of it on kimmy in 13 min (802 s, one core; tm `cert-check-c4`, 20261101-182226). The GAP records were not kept in the runs above: 32 were recovered from the Modal volumes (31 groups $M$ and $Q$ of P6), the other 128 rebuilt (`scripts/c4_rebuild_gap.sh`; T from the parameters logged by `gap/c4_rand.g`, `gap/c4_T_fromlog.g`), and the checker reproduces every index and count of the original outputs with them; for the rebuilt records only these basis-independent numbers are claimed, for the kept ones also the membership of every test (except Rk14, …, Rk56 at $n = 6$, whose run used an earlier record without $w$). Not re-checked: the indices modulo $2^{30}$, $2^{20}$. Trusted: the GAP records (GAP 4.14 + nq; all relators; the `test_coeffs` are the elements $(\mathrm{ord}_t/2)\,b_t$ of an snf basis $b_t$ of $\gamma_c$, all of them, and the `kgen_coeffs` that basis — the checker verifies the first against the second where $\gamma_4$ is finite, 51 of the 80 groups $M$), Lemma 3, and on paper Lemma 1, Proposition 4, Corollary 2.

**Modal spend for §6.8:** $0.79 (one CPU container for GAP, 13 min, $0.07; two A10G runs at $n = 7$, 15 min each, $0.72), app `math43-kfree`, stopped. The 10 R2 bases at $n = 7$: $4.03 (CPU container for GAP, 8 cores / 48 GiB, 35 min, $0.44, app `math43-r2n7-gap`; ten A10G runs, $3.59, app `math43-r2n7`, stopped). The last 5 bases (R, Rd46f15) at $n = 7$: $1.87 (CPU container for GAP, 5 cores / 28 GiB, 15 min, $0.12, app `math43-c4rest-gap`; two A10G calls, 28 + 43 min, $1.75, app `math43-c4rest`; both stopped). Total for §6.8: $6.69.

References for §6.8 († = not checked against the source): [MP09] R. Mikhailov, I. B. S. Passi, LNM 1952 (§7) — for $D_3 = \gamma_3$ †; S. A. Jennings, *The group ring of a class of infinite nilpotent groups*, Canad. J. Math. 7 (1955) — rational dimension subgroups †; K. W. Gruenberg, *Residual properties of infinite soluble groups*, Proc. London Math. Soc. (3) 7 (1957) — residual $p$-finiteness of finitely generated nilpotent groups †.

### 6.9. D5-7: the ring $L'_2$ on four generators, checked by computer; its group analogue

**The ring** (§6.7, last paragraph). $L'_N$: the free metabelian Lie ring on $x_1..x_4$, Hall order $x_3 < x_1 < x_2 < x_4$, $t = [x_4, x_3, \dots, x_3]$ ($N$ letters $x_3$), $c_i = [t, x_i]$, $u_i = [c_i, x_i]$, the relators $r_1 = 4x_1 + 2c_3 + c_2$, $r_2 = 16x_2 + 4c_3 - c_1$, $r_3 = 64x_3 - 4c_2 - 2c_1$, (ii) $\gamma_{N+4} = 0$, (iii) every Hall basis element of weight $N + 3$ other than $u_1, u_2, u_3$ is 0, (iv) $u_1 = 16u_3$, $u_2 = 4u_3$, $64u_3 = 0$; $a = 32[x_1, x_2] + 64[x_1, x_3] + 128[x_2, x_3]$. Claimed: $D_{N+4} = \cdots = D_{2N+4} = \langle a\rangle \cong \mathbb{Z}/2$, $D_{2N+5} = 0$.

**Method** (`py/lie_L2.py`, `c/lie_howell.c`). $U(L) = \mathbb{Z}\langle X\rangle/(\text{relators})$, so $U(L)/\varpi^n = A_n/J_n$ with $J_n = \mathrm{span}\{u\rho v\}$ truncated at length $n$, and a Lie element $\ell$ lies in $D_n(L)$ iff $\ell \in J_n$ (PBW: $L \to U(L)$ is injective). $L = W/\bar R$, $W$ the span of the left-normed brackets of weight $\le N + 3$ in $A_n$, $\bar R$ the span of $[\rho, x_{j_1}, \dots, x_{j_k}]$ truncated at weight $N + 3$; so $D_n(L) = (W \cap J_n)/\bar R$. *Grading:* with $\deg x_1 = \deg x_2 = \deg x_3 = 1$, $\deg x_4 = -N$ every relator is homogeneous ($\deg c_i = \deg x_i$), so everything splits into components; $a$ has degree 2. For $n \le 6$ all components are computed exactly over $\mathbb{Z}$ (HNF in python-flint), for $n = 7, 8$ the component of $a$; for $n = 9$ the component of $a$ (20 826 columns, 172 674 sparse rows $u\rho v$) by a Howell form modulo $2^E$ built by row insertion in C (2 min; the Howell property is checked by reducing $2^{E-v}P$ for every pivot). "$\notin$" modulo $2^E$ is exact; "$\in$" is certified by the index being the same for several $E$ with no free columns. The metabelian relators are $[U, V]$ for all left-normed $U, V$ of weights $\ge 2$ with $w(U) + w(V) \le N + 3$, plus all left-normed brackets of weight $N + 4$. *(The brackets*

$[[x_i,x_j],[x_k,x_l]]$ *alone do not generate* $F''$ *as a Lie ideal: with them* $x_2[c_2,t]\notin J_9$, *i.e.* $4c_1\neq 0$ *and (3.9) failed at* $n=9$. *This was a bug of the first version of the script; the group library* `gap/plateau5_lib.g` *already does it right.)*

**Validation on [IMP26, Theorem A],** $N=1$ (`--var orig`, five generators, a $\mathbb{Z}^2$-grading): $D_5(L_1)=\langle a\rangle\cong\mathbb{Z}/2$ (exact, all 25 components), $a\in D_6$ (exact), $a\notin D_7$ (modulo $2^{16}$ and $2^{32}$), $a\equiv c_1c_3-c_2^2 \bmod \varpi^7$.

$L_2'$ [computer]:

| $n$ | how | result |
|---|---|---|
| 6 | exact, all 15 components | $D_6=\langle a\rangle\cong\mathbb{Z}/2$ ($a\neq 0$, $2a=0$ in $L$) |
| 7 | exact, component of $a$ | $D_7=\langle a\rangle$ |
| 8 | exact (component of $a$) and modulo $2^8,2^{16},2^{32}$ (index 745 for all three, no free columns) | $a\in D_8$, so $D_8=\langle a\rangle$ |
| 9 | modulo $2^8,2^{12},2^{20},2^{32}$ (index 14 050 for all, no free columns) | $a\notin\varpi^9$, so $D_9=0$ |

($D_7,D_8,D_9\subseteq D_6=\langle a\rangle$, so the component of $a$ suffices from $n=7$ on.) Also at $n=9$: $4c_1=16c_2=64c_3=0$ (3.8) and $a\equiv c_1c_3-c_2^2$ (3.9), while $c_1c_3-c_2^2\notin\varpi^9$. So the claim of §6.7 holds: $D_6(L_2')=D_7=D_8=\langle a\rangle\cong\mathbb{Z}/2$, $D_9=0$, **on four generators, class 5.** Theorem C gives $D_9=0$ anyway; the computation agrees with it.

**The plateau is forced by the grading** [on paper]. In the component of $a$ (degree 2) a word with $k$ letters $x_4$ has $2+Nk$ other letters, so its length is $2+(N+1)k\in\{2,N+3,2N+4,\dots\}$; the same holds for $L_N$ of [IMP26] with the grading $\deg x_5=(0,1)$, $\deg x_4=(0,-N)$, $\deg x_{1,2,3}=(1,0)$. Since $\varpi^n$ is spanned by the words of length $\geq n$, $\varpi^{N+4}\cap U(L)_2=\varpi^{2N+4}\cap U(L)_2$, hence $D_{N+4}\cap L_2=D_{2N+4}\cap L_2$: once $D_{N+4}\subseteq L_2$, the length $N+1$ of the plateau is the gap between the admissible lengths $N+3$ and $2N+4$. (Computed: for $N=2$ the component of $a$ in $A_6,A_7,A_8$ is the same 414 columns and 1 791 rows.) A group ring has no such grading: $\mu(x^E)-1=\sum_k\binom{E}{k}a^k$ mixes all lengths.

**The group analogue** [computer]. The literal translation `L2g` (`gap/plateau5_cands.g`, family `I5cand`, $d=4$): $t=[x_4,x_3,x_3]$, $d_l=[t,x_l]$,

$$x_1^4d_2d_3^2,\quad x_2^{16}d_1^{-1}d_3^4,\quad x_3^{64}d_1^{-2}d_2^{-4},\quad [d_1,x_1][d_3,x_3]^{-16},\quad [d_2,x_2][d_3,x_3]^{-4},\quad [d_3,x_3]^{64},\quad [d_l,x_m]\ (l\neq m),\ [d_4,x_4],$$

$w=[x_1,x_2^{32}][x_1,x_3^{64}][x_2,x_3^{128}]$ (the image of $a$). Then $G_{\mathrm{ab}}=\mathbb{Z}\oplus\mathbb{Z}/4\oplus\mathbb{Z}/16\oplus\mathbb{Z}/64$ as for $L_2'$, $\gamma_5(G)\cong\mathbb{Z}/64=\langle u_3\rangle$, $u_3=[d_3,x_3]$, $w=u_3^{32}$; GAP 13 min. Tests of $u_3^{2^j}$ (CPU, `howell_gpu.py`, certified): $D_5\cap\gamma_5=\gamma_5$, $D_6\cap\gamma_5=\langle w\rangle$, $D_7\cap\gamma_5=1$. **The group analogue stops at** $D_6$**, two levels below the ring.** With $L_{ij}$ doubled or quadrupled, $w=1$ already in $M$.

**Where it breaks** (the pairing lemma of §6.7, made sharp). In the ring $e_ix_i=-C_i\in\gamma_4\subseteq\varpi^4$, and (3.9) is $a=\sum_i x_ig_i=\sum_i(e_ix_i)\zeta_i\in\varpi^{4+4}$ with $g_i=\sum_j L_{ij}x_j=e_i\zeta_i$. In a group, $x_i^{E_i}\in\gamma_4$ gives only $2^jE_i(x_i-1)\in\Delta^{2+j}$ ($j\leq 2$), and this is sharp when $\bar{x}_i$ spans a cyclic direct summand $\mathbb{Z}/o$ of $G_{\mathrm{ab}}$ (as in `L2g`, $o=E_i$): mapping to $C_o$, the least $F$ with $F(x-1)\in\Delta^k(C_o)$ is $2^{k-2}o$ [computer, `scripts/cyclic_pairing_check.py`, $o=2,\dots,256$, $k=2..5$; on paper for $k\leq 4$: $o(x-1)\equiv-\binom{o}{2}(x-1)^2\not\equiv 0$ and $2o(x-1)\equiv\frac{o(o^2-1)}{6}(x-1)^3\not\equiv 0$ in $\Delta^k/\Delta^{k+1}(C_o)\cong\mathbb{Z}/o$]. So the term $(e_ix_i)\zeta_i$ becomes $\Delta^{2+4}=\Delta^6$ — exactly where `L2g` stops; $\Delta^8$ needs $g_i\in 4E_iD$ (multiplier 4, $D=\gamma_4/\gamma_5$). On the other side, $u_{lm}=[d_l,x_m]$ has order dividing the order $o_m$ of $\bar{x}_m$ in $G_{\mathrm{ab}}$ ($[d_l,x_m^{o_m}]\in[\gamma_4,\gamma_2]=1$ in class 5), and $w$ is a combination of the $u_{lm}$ with coefficients $L_{ij}/E_j$; the multiplier pushes $L_{ij}$ up by the same factor. In the bilinear model of the top two layers for the coefficient pattern of $L_2'$ (`scripts/lie_L2_pairing_screen.py`: $E_i=2^{a_i}$, $a_i\leq 12$, all admissible $L_{ij}$): multiplier 1 has 84 solutions with $w\neq 0$ (the cheapest is $L_2'$ itself, $w=8u_{22}=a$), multipliers 2 and 4 have none [computer, model]. (3.12) gets multiplier 2 at class 4 only through the fifth generator $x_5$ with a large order, which carries $w$ (§6.7); the 4-generator attempts of §6.7 (class 4, multiplier 2) all failed, and the 80 cheapest multiplier-4 scalings of (3.12) (five generators) have $w=1$. [Conjecture] a metabelian group of class 5 with $D_8\neq 1$, if it exists, needs a mechanism other than the substitution $e_ix_i\mapsto x_i^{E_i}$ (for instance a generator carrying $w$ whose order is not tied to the paired ones, as $x_5$ in (3.12)); on four generators we have none.

$D_9$ **test.** Not run: no group with $w\in D_8$ was found. Cost if one appears ($d=4$, $n=9$, $M=87\,380$): the dense form is 30.5 GB (uint32), the closure peak $\approx 1.64\times$ that $\approx 50$ GB, so an A100 80 GB; the $d=5$, $n=8$ run ($M=97\,655$) took 30 min, so about 30 min, $\approx 1.3$ USD at 2.5 USD/h, plus the GAP build on a CPU container. Modal spend for §6.9: 0 (everything local).

**Rerun.**

```
# exact D_6, all components (needs python-flint; 20 min)
python3 py/lie_L2.py exact --var prime --N 2 --n 6 --full-upto 6
# the component of a:      ... --n 7 8
# validation:              ... --var orig --N 1 --n 5 --full-upto 5
```

```
# n=9
python3 py/lie_L2.py rows --var prime --N 2 --n 9 --out F.bin
cc -O3 -o c/build/lie_howell c/lie_howell.c
c/build/lie_howell F.bin F.bin.tests 12 [1]
# the identities
python3 scripts/lie_L2_identities.py prime 2 9 T && c/build/lie_howell F.bin T 12
# the group, with the spec in results/L2/scan_L2g.jsonl (the JSON of G is gpu/data/plateau5/L2g.jsonl.gz)
python3 scripts/plateau5_scan.py LOG 'L2g|I5rec(...)|5,6,7'
```

Raw outputs: `results/L2/`.

### 6.10. D5-7, class 5: all $K \subseteq \gamma_5(M)$ at once for the shifted plateau

The test of §6.8(b) (Proposition 4), moved to the class-5 group of §6.7 (`plateau5_c5i5`: five generators, $t = [x_4, x_3, x_3]$, $d_i = [t, x_i]$, the nine relators of (3.12); there $G = M/\ker\varphi$ with $\gamma_5(G) \cong C_{2^{14}}$ and $w \in D_7(G) \setminus D_8(G)$, certified). Here $\Gamma$ is that presentation, $M = \Gamma/\gamma_6(\Gamma)\Gamma''$, $A_5 = \gamma_5(M)$, $A = \mathbb{Z}[M]$, $n \le 10$. Data: `results/c5/kfree/`.

**Lemma 5** [on paper] (Lemma 3 for $c = 5$). (i) $\psi : A_5 \to \Delta^5/\Delta^n$, $k \mapsto k - 1$, is a homomorphism for $n \le 10$: $kk' - 1 = (k-1) + (k'-1) + (k-1)(k'-1)$ and $(k-1)(k'-1) \in \Delta^{10}$. (ii) For $K \subseteq A_5$: $(K-1)A + \Delta^n = \Delta^n + \sum_k \psi(k)R$, $R = \mathrm{span}\{1\} \cup \{(x_{j_1} - 1)\cdots(x_{j_l} - 1) : 1 \le l \le n-6\}$, $k$ over generators of $K$ ($K$ is central, $\psi(k)\Delta^{n-5} \subseteq \Delta^n$, and $\Delta/\Delta^{n-5}$ is spanned by these monomials). The proof is that of Lemma 3 with $4 \to 5$.

**Proposition 6** [on paper] (Proposition 4 for $c = 5$). Let $U_n = J_n + \Lambda(A_5 \otimes R_+)$, $R_+$ the monomials of degree $1..n-6$, $\bar\psi_n : A_5 \to A/U_n$. If $\bar\psi_n$ is injective, then $D_n(M/K) \cap \gamma_5(M)/K = 1$ for every $K \subseteq A_5$. Conversely every $v \in D_n(M/K) \cap A_5$, $v \notin K$, gives $1 \ne vk^{-1} \in \ker\bar\psi_n$ for some $k \in K$, and $D_n(M) \cap A_5 \subseteq \ker\bar\psi_n$. Further: $\ker\bar\psi_n$ is torsion ($z \in \ker\bar\psi_n$ gives $z - 1 \in \Delta^6$, so $z \in D_6(M) \subseteq \sqrt{\gamma_6(M)} = \tau(M)$, Jennings), so it is enough to test the elements of order 2 of $\tau(A_5)$; $\ker\bar\psi_{n+1} \subseteq \ker\bar\psi_n$ ($n+1 \le 10$); injectivity found modulo $2^{32}$ is exact. Proof: as for Proposition 4. For $n = 6$, $R_+ = \varnothing$ and $U_6 = J_6$: the test is just $D_6(M) \cap \gamma_5$.

**Removing** $\cap\gamma_5$ [on paper]. For $K \subseteq \gamma_5(M)$, $(M/K)/\gamma_5(M/K) = M/\gamma_5(M) = \Gamma/\gamma_5(\Gamma)\Gamma'' =: Q$, a class-4 group that depends neither on $K$ nor on $\varphi$ (also $G/\gamma_5(G) = Q$). The image of $D_n(M/K)$ in $Q$ lies in $D_n(Q)$. So

$$D_n(Q) = 1 \text{ and } \bar\psi_n \text{ injective} \implies D_n(M/K) = 1 \text{ for every } K \subseteq \gamma_5(M).$$

$D_8(Q) = 1$ was **not** known: §6.8(a) is about the class-3 quotients of the class-4 bases, and §6.7 tested only $D_n(G) \cap \gamma_5$. Since $D_8(Q) \subseteq D_5(Q)$, it is enough to have $D_5(Q) = 1$, and by Corollary 2 (with $Q_3 = Q/\gamma_4(Q) = \Gamma/\gamma_4(\Gamma)\Gamma''$) that follows from $D_5(Q_3) = 1$ (Lemma 1) and $D_5(Q) \cap \gamma_4(Q) = 1$ (the map $\gamma_4(Q) \to \Delta^4/\Delta^5$ is additive and its kernel is torsion; the elements of order 2 must stay independent modulo $J_5(Q)$). A separate test, but a cheap one ($n = 5$, $\dim A_5 = 780$).

**Computation** [computer]. `scripts/c4_q3.py gapin` now also reads records of `gap/plateau5_lib.g` (it drops the 1 250 relators of $\ker\varphi$ and the 6 250 weight-6 commutators, keeping the 9 relators of $\Gamma$ and the 545 metabelian ones); `gap/c4_q3.g` with `Q3_C` $= 3, 4, 5$ builds $Q_3$, $Q$, $M$ (nq on a Modal CPU container, 2 cores; with the weight-6 commutators passed to nq the class-5 run did not finish in 30 min and was stopped, without them, `Q3_NQTOP := false`, it took 33 min, almost all in GAP after nq); `howell_gpu.py batch --kfree` now takes the degrees of $R_+$ from the record's class ($1..n-c-1$).

| group | $\gamma_c$ (torsion-free ranks $h$) | elements of order 2 | $n$ | index $J_n$ ($2^{32}/2^{30}/2^{20}$), certificate |
|---|---|---|---|---|
| $Q_3$, $c = 3$ | $C_8^{20}C_{64}^{12}C_{1024}^{6}C_{16384}^{2}$, $h = (1,0,0)$ | 40 | 4; 5 | 983/977/947; 3 308/3 300/3 260, $r_n = 3, 4,$ |
| $Q$, $c = 4$ | $C_8^{60}C_{64}^{30}C_{1024}^{12}C_{16384}^{3}$, $h = (1,0,0,0)$ | 105 | 5; 6 | 3 830/3 822/3 782; 13 704/13 694/13 644, |
| $M$, $c = 5$ | $C_4^{2}C_8^{137}C_{16}C_{32}C_{64}^{57}C_{256}C_{512}C_{1024}^{18}C_{16384}^{4}$, $h = (1,0,0,0,0)$ | 222 | 6 | 14 722/14 712/14 662, $r_6 = 5$, holds |
| | | | 7 | 54 097/54 085/54 025, $r_7 = 6$, holds |

(For $M$ at $n = 7$: $U_7$ adds 1 110 rows $\psi(b_t)a_j$ for the 222 generators $b_t$ of $\gamma_5(M)$, index $U_7 = 49\,496$.) Runs: $Q_3$, $Q$, $M$ at $n = 6$ on the local CPU (6 s, 4.5 min, 4 min); $M$ at $n = 7$ on a Modal A10G (194 s, GPU peak 5.2 GiB, ref `0ac93a8`, a WIP commit of this branch with the same code); a CPU run of $n = 7$ gave the same ranks and indices for rounds 0–3 and was then stopped (`m5_n7_cpu_partial.log`). Lines: `results/c5/kfree/{q3,q4}.out.jsonl`, `m5_n6.out.jsonl`, `m5_n7_a10g.out.jsonl`.

**What this gives.**

- [computer + on paper] $D_4(Q_3) = 1$ and $D_5(Q) = 1$, so $D_8(Q) = 1$. With §6.7 ($D_8(G) \cap \gamma_5(G) = 1$) this closes the claim of §6.7 that the class-5 group $G$ has $D_8(G) = 1$: $D_8(G)$ maps into $D_8(Q) = 1$, so $D_8(G) \subseteq \gamma_5(G)$. Also $D_7(G) \subseteq \gamma_5(G)$ and $D_n(M/K) \subseteq \gamma_5(M)/K$ for every $K$ and $n \geq 5$.
- [computer] At $n = 6$ and $n = 7$ the kernel of $\bar{\psi}_n$ has exactly one independent element of order 2, and $w \in J_n(M)$ (with $w_{12}, w_{13}, w_{23}$; $u_3 \notin J_n$): $w \in D_7(M)$ for $M$ itself, not only for $G$. So $\ker \bar{\psi}_7$ is a nontrivial cyclic 2-group containing $w$, and $D_7(M) \cap \gamma_5(M)$ is a nontrivial subgroup of it. The order of $w$ in $M$ was not computed (in $G$ it is 2); the element of order 2 of the kernel is $z = w^{|w|/2}$.
- [on paper] Hence the whole question "is there $K \subseteq \gamma_5(M)$ with $D_8(M/K) \neq 1$" reduces to one bit: $\ker \bar{\psi}_8 \subseteq \ker \bar{\psi}_7$ is cyclic, so $\bar{\psi}_8$ is injective iff $z \notin U_8$, i.e. iff `socle_bits` $= 222$ at $n = 8$. If yes: $D_8(M/K) = 1$ for every $K \subseteq \gamma_5(M)$ — no central modification of the shifted plateau has $D_8 \neq 1$. If no: $z \in D_8(M/K)$ is not yet implied (the test is a filter, as in §6.8), and the next step is the representation $\psi(z) = \sum_j \psi(k_j) m_j$ modulo $J_8$ with $z \notin \langle k_j \rangle$ (the augmented Howell form of §6.8, not implemented).

$n = 8$ [computer] (Modal A100-SXM4-80GB, ref `ab3023d`, `gpu/modal_batch.py spawn --gpu a100 ... -- --n 8 --kfree --verbose --group 1`, app `math43-c5kfree-n8`; 2 622 s in all, of which the closure is 7 rounds and 1 881 s; GPU peak 61.0 GiB). $J_8(M)$: 6 804 relator rows, rank 97 648, free columns $7 = r_8$ for $h = (1, 0, 0, 0, 0)$, index 195 779 modulo $2^{32}$, 195 765 modulo $2^{30}$, 195 695 modulo $2^{20}$: differences $14 = 2r_8$ and $84 = 12r_8$, the certificate holds. $w, u_3, w_{12}, w_{13}, w_{23} \notin J_8(M)$. $U_8$ adds 6 660 rows $\psi(b_t)a_j$, $\psi(b_t)a_j a_{j'}$, index 170 084; the 222 elements of order 2 of $\gamma_5(M)$ generate a subgroup of order $2^{222}$ modulo $U_8$: $\bar{\psi}_8$ **is injective** (`socle_bits` $= 222$; exact without the certificate, which holds anyway). Line: `results/c5/kfree/m5_n8_a100.out.jsonl`.

**Conclusion** [computer + on paper, Proposition 6, $D_8(Q) = 1$ above]. For every $K \subseteq \gamma_5(M)$, $M = \Gamma/\gamma_6(\Gamma)\Gamma''$ with $\Gamma$ the shifted plateau (`plateau5_c5i5`): $D_8(M/K) = 1$. So no metabelian group of class $\leq 5$ of the form $\Gamma/N$ with $\gamma_6(\Gamma)\Gamma'' \subseteq N \subseteq \gamma_5(\Gamma)\gamma_6(\Gamma)\Gamma''$ has $D_8 \neq 1$ — all central modifications of the group of §6.7 at once, every $\varphi$ included; in particular $D_8(M) = 1$, while $w \in D_7(M)$. The candidate $z$ is gone: $\ker \bar{\psi}_8 = 1$. For $D_8 \neq 1$ at class 5 this $\Gamma$ is exhausted; a different presentation is needed (cf. §6.9).

**[computer, certified]** (D5-21, 01.11.2026). The socle counts, the indices modulo $2^{32}$ and the memberships of this subsection are confirmed by `cert/check_in.c`, which shares no code with `howell_gpu.py` (`results/cert/q{3,4,5}_n*_E32.claim`, `cert/README.md`). It builds $J_n$ by its own sparse closure; for $M$ it inserts the rows $\psi(b_t)m$ of $U_n$; then it inserts the elements of order 2, checks the Howell property after each step and that $2\psi(s)$ lies in $U_n$ (resp. $J_n$) $+ 2^{32}A_n^+$ for each of them, and takes the socle count as the index difference. $Q_3$: 40 of 40 modulo $J_4, J_5$; $Q$: 105 of 105 modulo $J_5, J_6$; $M$: **222 of 222 modulo** $U_8 + 2^{32}A_8^+$, so $\bar{\psi}_8$ is injective (a full count is exact without the certificate), and 221 of 222 at $n = 6, 7$ (modulo $U_n + 2^{32}A_n^+$); $w, w_{12}, w_{13}, w_{23} \in J_6, J_7$ (with the exactness check of §6.4 at $E = 32$), $\notin J_8$; indices of $J_n$ and $U_n$ as above. $M$ at $n = 8$: 100 s and 0.7 GB on one core of kimmy (the A100 run: 2 622 s). Not re-checked: the indices modulo $2^{30}$, $2^{20}$. Trusted: the records of `gap/c4_q3.g` — all relators; `kgen_coeffs` a basis of $\gamma_c$ with the orders `ord` (the checker verifies that the socle rows are (ord/2) times these rows, so they are a basis of the elements of order 2); for the memberships, $h$ (Quillen) and that $w, \ldots$ have 2-power order (they lie in the finite $\gamma_5(M)$); Lemma 5; and on paper Proposition 6, Lemma 1, Corollary 2 and Jennings' theorem.

**Modal spend for §6.10:** \$2.56: \$0.40 for $n \leq 7$ (GAP on a CPU container: the stopped first attempt \$0.12 and the run \$0.19, app `math43-c5kfree-gap`; one A10G run at $n = 7$, \$0.08, app `math43-c5kfree-n7`) and \$2.16 for $n = 8$ (A100-80GB, 44.5 min with the container of `modal_batch.py`, 4 CPU and 32 GiB, about 2.9 USD/h; app `math43-c5kfree-n8`). All stopped.

### 6.11. D5-9: what makes (3.12) work, and what a class-5 presentation would need

The question: find a metabelian group of class 5 with $D_8 \neq 1$, or say why the approach of §6.7–6.10 cannot give one. Here the mechanism of (3.12) is split into three parts that can be tested separately, by an exact identity in the group ring (Lemma 7, checked by computer); each part is measured in the groups we have (c4a = (3.12) with $x_6 \to x_3$, its version without $x_5$, $L_2'$'s literal group `L2g`, the shifted plateau `c5i5`). Data: `results/c5/presentation/`.

**Summary.**

- [on paper] $w - 1 \equiv S - XV - \sum_{i<j} B_{ij} \pmod{\Delta^{2k}}$ (Lemma 7): $S$ is the "ring part" (the pairing), $XV$ a commutator correction, $B_{ij}$ the binomial terms of the powers $x_j^{A_{ij}}$. In a Lie ring only $S$ is there.
- [computer] In (3.12) the fifth generator $x_5$ is **not** needed for the pairing: the exact pairing $g_i \in (\gamma_3)^{2E_i}$ holds with and without $x_5$, and $S \in \Delta^6$ in both. It is needed for the binomial error: without $x_5$ (or without its relator $x_5^{1024} d_2^{16}$) $B_{12} \notin \Delta^6$, more precisely $128(x_2 - 1)^2(x_1 - 1) \notin \Delta^6$; with it $B_{12} \in \Delta^6$ and $B_{13} + B_{23} \in \Delta^6$ (neither alone).

- [on paper] Two obstructions at class 5, both of size $c-3$ levels: (i) the pairing needs the multiplier $2^{k-2}=4$ (§6.7, §6.9); (ii) for the pair that carries $w$ ($A_{ij}<E_iE_j$ is forced, Lemma 9) the pairing lemma certifies the binomial error only up to $\Delta^{c+1}$ (Lemma 10), and $\Delta^{2c-2}$ needs $c-3$ more levels from other relators. Observed [computer]: without a carrier $w$ reaches exactly $D_{c+1}$ (c4b: $D_5$; `L2g`: $D_6$), with one carrier $D_{c+2}$ (c4a: $D_6$; `c5i5`: $D_7$).
- [computer] Three new families fail: the "$(c-2,c)$ split" ($w=1$ in all 62 groups built), the root extensions $x_i\mapsto y_i^m$ ($w=1$), and the equal-order solutions of the bilinear model (3 of 4 have $w=1$; in the fourth all $g_i=1$ and $w\in D_5\setminus D_6$).
- [conjecture] $D_8\neq 1$ at class 5 needs two independent carrier mechanisms, i.e. at least six generators. A dense test of such a group at $n=8$ is out of reach ($\dim A_8=335\,923$ for $d=6$).

**(a) The decomposition** Notation. $G$ any group, $\Delta$ its augmentation ideal, $X_i=x_i-1$ for $x_1,x_2,x_3\in G$; $A$ an antisymmetric integer matrix with $h_{ij}:=x_j^{A_{ij}}\in\gamma_k(G)$ ($i\neq j$); $g_i=\prod_{j\neq i}h_{ij}$, $u_{ij}=[x_i,h_{ij}]$ ($[a,b]=a^{-1}b^{-1}ab$), $v_i=\prod_{j>i}u_{ij}$, $w=\prod_{i<j}u_{ij}=[x_1,x_2^{A_{12}}][x_1,x_3^{A_{13}}][x_2,x_3^{A_{23}}]$ (the $w$ of (3.12), of §6.7, of `L2g`), and

$$S=\sum_i X_i(g_i-1),\qquad XV=\sum_i X_i(v_i-1),\qquad B_{ij}=\sum_{m\geq 2}\Big[\tbinom{A_{ij}}{m}X_j^mX_i+\tbinom{-A_{ij}}{m}X_jX_i^m\Big].$$

**Lemma 7** [on paper]. $w-1\equiv S-XV-\sum_{i<j}B_{ij}\pmod{\Delta^{2k}}$. *Proof.* $(h-1)(h'-1)\in\Delta^{2k}$ for $h,h'\in\gamma_k$, so $g_i-1\equiv\sum_j H_{ij}$ with $H_{ij}=h_{ij}-1$. From $x_ih=hx_i[x_i,h]$: $X_iH_{ij}-H_{ij}X_i=hx_i(u_{ij}-1)=(u_{ij}-1)+(X_i+H_{ij}+H_{ij}X_i)(u_{ij}-1)$, and $u_{ij}\in\gamma_{k+1}$, so $X_iH_{ij}\equiv H_{ij}X_i+(u_{ij}-1)+X_i(u_{ij}-1)$. Next $H_{ij}X_i+X_jH_{ji}=\sum_{m\geq1}\binom{A}{m}X_j^mX_i+\sum_{m\geq1}\binom{-A}{m}X_jX_i^m$ ($A=A_{ij}$, $H_{ji}=x_i^{-A}-1$; for a negative exponent the binomial series is finite modulo $\Delta^{2k}$), and the terms $m=1$ cancel: this is $B_{ij}$. Summing over $i<j$: $S\equiv\sum_{i<j}(u_{ij}-1)+XV+\sum B_{ij}$; finally $\sum_{i<j}(u_{ij}-1)\equiv w-1$ since $(u-1)(u'-1)\in\Delta^{2k+2}$, and $\sum_{j>i}X_i(u_{ij}-1)\equiv X_i(v_i-1)$. □

No hypothesis on $G$ beyond $h_{ij}\in\gamma_k$ (not even metabelian). In a Lie ring the analogue is exact with $XV=B=0$: $a=\sum_{i,j}A_{ij}x_ix_j=\sum_i x_ig_i$ [IMP26, (3.9)]. So $w\in D_{2k}$ iff $S-XV-\sum B_{ij}\in\Delta^{2k}$, and the three parts are the three places where a group can lose levels against a ring.

**Lemma 8** [on paper] (sufficient conditions). Let $x_i^{E_i}\in\gamma_k$, $E_i=2^{a_i}\geq 4$, $k\leq 4$ (so the pairing lemma of §6.7 holds: $2^sE_iX_i\in\Delta^{2+s}$ for $2+s\leq k$). (i) If $g_i=\zeta^Fs^{F'}$ with $\zeta\in\gamma_k$, $s\in\gamma_{k+1}$, $F=2^{k-2}E_i$, $F'=2^{k-3}E_i$, then $X_i(g_i-1)\in\Delta^{2k}$. (ii) If $v_i\in(\gamma_{k+1})^{2^{k-3}E_i}$, then $X_i(v_i-1)\in\Delta^{2k}$. *Proof.* $\zeta^F-1\equiv F(\zeta-1)$ modulo $\Delta^{2k}$, and $X_i\cdot F(\zeta-1)=(FX_i)(\zeta-1)\in\Delta^{k+k}$; likewise $X_i(s^{F'}-1)\equiv(F'X_i)(s-1)\in\Delta^{(k-1)+(k+1)}$; the cross term $(\zeta^F-1)(s^{F'}-1)\in\Delta^{2k+1}$. (ii) is the second half of (i). □

This is the "exact pairing": the multiplier is $2^{k-2}$ on the $\gamma_k$-part of $g_i$ (2 for (3.12), 4 at class 5) and $2^{k-3}$ on its $\gamma_{k+1}$-part. Condition (ii) holds as soon as the top layer has elements of large order (in $G$ of (3.12) and of §6.7, $\gamma_c(G)=C_{2^{14}}$).

**Lemma 9** [on paper]. Let $G$ be metabelian of class $c$, $k=c-1$, $x_i^{E_i}=P_i\in\gamma_k$, and $E_iE_j\mid A_{ij}$. Then $u_{ij}=1$. Hence $w\neq 1$ forces $E_iE_j\nmid A_{ij}$ for some $i<j$ (for powers of two: $A_{ij}<E_iE_j$). *Proof.* $u_{ij}=[x_i,P_j^q]=[x_i,P_j]^q$, $q=A_{ij}/E_j$, since $[x_i,P_j]\in\gamma_c$ is central; and $[x_i,P_j]^{E_i}=[x_i^{E_i},P_j]=[P_i,P_j]=1$ ($\gamma_k\subseteq G'$ is abelian). □

**(b) Which part fails: measured** `scripts/c5_lintests.py` writes $S$, $\mathrm{Err}=S-(w-1)$, $XV$, $B_{12},B_{13},B_{23}$, $\sum B$, $XV+\sum B$ and $\mathrm{chk}=\mathrm{Err}-XV-\sum B$ as integer combinations $\sum_k c_k(w_k-1)$ of words (field `lintests` of a record); `gpu/howell_gpu.py` (new: rows $\sum_k c_k(\mu(w_k)-1)$ of such combinations, reported after the ordinary tests) gives their membership in $J_n$, i.e. in $\Delta^n$; `scripts/c5_lintests_extra.py` adds any polynomial in the $X_i$. All runs local (CPU), all certificates hold (free columns $=r_n$, index differences $2r_n$, $12r_n$).

| group | $c,k$ | $n$ | $w$ | $S$ | $XV$ | $B_{12}$ | $B_{13}$ | $B_{23}$ | $\sum B$ | chk |
|---|---|---|---|---|---|---|---|---|---|---|
| c4a: (3.12), $x_6\to x_3$ (`plateau5_c4i5`) | 4, 3 | 5 | ∈ | ∈ | ∈ | ∈ | ∈ | ∈ | ∈ | ∈ |
| | | 6 | ∈ | ∈ | ∈ | ∈ | ∉ | ∉ | ∈ | ∈ |
| | | 7 | ∉ | ∉ | ∈ | ∉ | ∉ | ∉ | ∉ | ∈ |
| c4b: c4a without $x_5$ | 4, 3 | 5 | ∈ | ∈ | ∈ | ∈ | ∈ | ∈ | ∈ | ∈ |
| | | 6 | ∉ | ∈ | ∈ | ∉ | ∉ | ∉ | ∉ | ∈ |
| c4a without $x_5^{1024}d_2^{16}$ | 4, 3 | 6 | ∉ | ∈ | ∈ | ∉ | ∉ | ∉ | ∉ | ∈ |
| `L2g` (§6.9) | 5, 4 | 6 | ∈ | ∈ | ∈ | ∈ | ∈ | ∈ | ∈ | ∈ |
| | | 7 | ∉ | ∉ | ∈ | ∉ | ∉ | ∉ | ∉ | ∈ |
| `c5i5` (§6.7) | 5, 4 | 8 | ∉ | ∉ | ∈ | ∉ | ∉ | ∉ | ∉ | ∈ |

chk $\in\Delta^n$ everywhere, as Lemma 7 says. And the pieces of $B_{12}\equiv 128(X_2X_1^2-X_2^2X_1)$ (mod $\Delta^6$, Lemma 10 below):

| group | $n$ | $128X_2X_1^2$ | $128X_2^2X_1$ |
|---|---|---|---|
| c4a | 6 | $\in$ | $\in$ |
| c4a | 7 | $\in$ | $\notin$ |
| c4b | 6 | $\in$ | $\notin$ |
| c4a without $x_5^{1024}d_2^{16}$ | 6 | $\in$ | $\notin$ |
| `c5i5` | 8 | $\in$ | $\notin$ |

(Sanity: $16X_1$, $128X_2$, $8X_1^2$, $64X_2^2$ are $\notin \Delta^6$ in both, as they must be.)

**The exact pairing** (`gap/c5_ep.g`, in $M = \Gamma/\gamma_{c+1}\Gamma''$ and in $G = M/\ker\varphi$; the largest $a$ with $g_i \in (\gamma_k)^{2^a}\gamma_c$, and whether the $\gamma_c$-part is right, Lemma 8(i)) [computer]:

| group | $g_1$ | $g_2$ | $g_3$ |
|---|---|---|---|
| c4a ($E = 8, 64, 16384$) | $(\gamma_3)^{16}$ exactly: multiplier 2 | $(\gamma_3)^{128}$ exactly: 2 | $g_3 = 1$ |
| c4b | the same | the same | $g_3 = 1$ |
| `c5i5` (class 5, $k = 4$) | $(\gamma_4)^{16}$, not $(\gamma_4)^{32}\gamma_5$: multiplier 2, need 4 | $(\gamma_4)^{128}$ only: 2, need 4 | $g_3 = 1$ |

(The same in $M$ and in $G$ for all three; $w$ has order 2 in $M$ in all three. Logs `ep_c4a.log`, `ep_c4b.log`, `ep_c5i5.log`.)

So in (3.12) the pairing does not see $x_5$ at all (the same multipliers, the same exactness, in $M$ and in $G$), and $S \in \Delta^6$ with or without it. **What $x_5$ does** [computer]: it puts $128X_2^2X_1 = 2E_2\,X_2^2X_1$ into $\Delta^6$ (the pairing lemma alone gives $\Delta^5$), hence $B_{12} \in \Delta^6$, and it makes $B_{13} + B_{23} \in \Delta^6$ although neither term is. Dropping only its power relator $x_5^{1024}d_2^{16}$ undoes the first effect. The lift is exactly one level: at $n = 7$, $128X_2^2X_1 \notin \Delta^7$ in c4a (CPU, 3.8 h on a shared machine, certificate holds; `c4a_lin2_n7.log`). (Which identity in $\mathbb{Z}[G]$ does it, was not found on paper.) `L2g` fails at $n = 7$ in $S$ (multiplier 1 where 4 is needed) and in every $B_{ij}$.

**(c) Level count: why a carrier buys one level** Call the *certified level* of $NX_{t_1}\cdots X_{t_m}$ ($N$ a power of two) the largest $\sum_r \ell_r$ over factorisations $N = \prod_r N_r$, where $\ell_r$ is the level that the pairing lemma gives to $N_rX_{t_r}$ (1 if $N_r < E_{t_r}$, else $\min(2 + \log_2(N_r/E_{t_r}), k)$). It is a lower bound for the true level; relators can do better.

**Lemma 10** [on paper]. Let $E_i \le E_j$, $A = A_{ij} = 2^\alpha < E_iE_j$ (the pair of Lemma 9). The certified level of $\binom{A}{2}X_j^2X_i$ is at most $k + 2 = c + 1$. *Proof.* $v_2\binom{A}{2} = \alpha - 1 \le a_i + a_j - 2$. Raising $X_j$ to level $\ge 2$ costs $\ge a_j$, and then $\le a_i - 2 < a_i$ is left: the other two factors stay at level 1, total $\le k + 2$. Otherwise only $X_i$ is raised, to $\le k$: total $\le k + 2$. □

The same count gives $S$ the certified level $k + \min(2 + \log_2 m, k)$ for multiplier $m$, and $XV$ the level $2k$ under Lemma 8(ii). So with the pairing lemma alone the decomposition certifies $w \in D_{c+1}$ at best, and $D_{2c-2}$ needs $c - 3$ further levels for the binomial term of the $w$-pair: one at class 4, two at class 5. The data fit exactly:

| group | $c$ | carrier | multiplier | certified ($S$; $B_w$) | observed |
|---|---|---|---|---|---|
| c4b, c4a without $x_5$'s relator | 4 | none | 2 | 6; 5 | $w \in D_5 \setminus D_6$ |
| c4a ((3.12)) | 4 | $x_5$ | 2 | 6; 5 (+1 by $x_5$, measured) | $w \in D_6$, $D_7(M/K) = 1$ (§6.8) |
| `L2g` | 5 | none | 1 | 6; 6 | $w \in D_6 \setminus D_7$ |
| `c5i5` (§6.7) | 5 | $x_5$ | 2 | 7; 6 | $w \in D_7 \setminus D_8$ |

[conjecture H] A carrier relation of the kind of $x_5^{1024}d_2^{16}$ lifts the binomial error of the $w$-pair by one level and not more; $D_8$ at class 5 then needs the multiplier 4 *and* two such mechanisms. Within five generators (the most a dense test at $n = 8$ can take) there is room for one.

**(d) What a class-5 presentation needs** With $k = c - 1 = 4$, by Lemmas 7–10 [on paper]:

- (R1) pairing: $g_i \in (\gamma_4)^{4E_i}(\gamma_5)^{2E_i}$ for every $i$, or a cancellation $S \equiv XV + \sum B$ modulo $\Delta^8$ not of this kind;
- (R2) $v_i \in (\gamma_5)^{2E_i}$: a top layer with elements of order $\gg E_i$ (a cyclic $\gamma_5(G)$ of large order, as in §6.7);
- (R3) the binomial error of the $w$-pair in $\Delta^8$: two levels above what the pairing lemma certifies;

- (R4) $w \neq 1$: $A_{ij} < E_iE_j$ for some pair (Lemma 9) — which is exactly what makes (R3) hard (Lemma 10), and in the $L'_2$ pattern also what makes (R1) impossible (§6.9: multipliers 2 and 4 have no solutions).

The bilinear model of §6.9 with *free* coefficients $p_{il}$ does have multiplier-4 solutions. [on paper] With only $d_2$ in the relators ($x_1^{E_1} = d_2^{p_{12}}$, $x_3^{E_3} = d_2^{p_{32}}$) the conditions reduce to $v_2(q_{21}p_{12}) = v_2(q_{23}p_{32}) < a_2$ and two inequalities that hold for $A_{13}$ large; with $E_1 = E_2 = E_3$ the multiplier does not enter at all (the $d_l$ have order $\leq E_l$, so $\gcd(4E_i, E_l) = E_l$). [computer, `scripts/c5_split_screen.py`] For equal orders $E \leq 2^6$, $p_{il} \in \{0, \pm 1, \pm 2\}$: 40 solutions, the cheapest $E = 4$, $A = (4, 8, 4)$, $p_{12} = \pm 2$, $p_{32} = \pm 2$. But the model's "$w \neq 0$" is not reliable: of four such groups built in GAP (`eq4a`, `eq4b`, `eq8a`, `eq8b`: $E = 4$ or 8 for all three, $P_1 = d_2^{\pm 2}$, $P_3 = d_2^2$, class 5, $t = [x_4, x_3, x_3]$) three have $w = 1$ in $M$, and in `eq8a` ($w$ of order 2) all $g_i = 1$ exactly — $S = 0$, $w$ is pure binomial error, and $w \in D_5 \setminus D_6$ (certified) [computer].

**(e) Other candidates, and why they fail**

1. **The** $(c-2, c)$ **split.** Pair at weight $c-2$ and put the second factor in the centre: if $x_i^{E_i} \in \gamma_{c-2}$ and $g_i \in (\gamma_c)^{2^{c-4}E_i}$, then $X_i(g_i - 1) \in \Delta^{2c-2}$ (as Lemma 8: $2^{c-4}E_iX_i \in \Delta^{c-2}$), with the multiplier 2 at class 5, as in (3.12). (Lemma 7 then holds only modulo $\Delta^{2c-4}$; the error terms are larger.) [on paper, sketch] In a Lie ring this cannot carry $D_{2c-2}$: the retraction of the proof of [IMP26, Theorem C] (onto $T$, vanishing on $Z(T) \cdot T$) kills $\sum_i (e_i x_i)\zeta_i$ with $\zeta_i$ central, while it fixes $a$; in the shifted family the $\gamma_c$-part of $g_i$ comes only from binomials ($E_l d_l = -\binom{E_l}{2}[d_l, x_l]$), so it is zero in the ring. [computer] The lab test at class 4 ((2,4) split: $d = [x_4, x_a]$ or $[x_a, x_b]$, $v$ of weight 4 in $x_3, x_4$, $x_i^{E_i} = d^{p_i}$, $d^o = v^s$; 60 random choices passing the one-$d$ model, `scripts/c5_split_gen.py 4 60 1` = `gap/c5_spl_c4_1.g`, `gap/c5_split.g`): $w = 1$ in $M$ in all 60. The explicit tries `lab24a`, `lab24b`: $w = 1$.
2. **Root extensions** $x_i \mapsto y_i^m$ (`gap/c5_roots.g`). For $x = y^m$ the multiplier drops [computer, `scripts/c5_cyclic_power_check.py`: in $\mathbb{Z}[C_N]$, $F(x-1) \in \Delta^k$ for $x = g^m$ of order $E$ needs $F/E = 2^{k-2}$ for $m = 1$, $(1, 1, 2, 2)$ for $m = 2$ and 1 for $m \geq 4$, $k = 2..5$]. But this is a change of coordinates, not a new mechanism: in the ring $a$ scales by $m^2$ and the orders only by $m$, so $a$ dies [on paper]. In GAP: `L1` (the group version of $L_1$, §6.6) has $w$ of order 2 in $M$, its square-root extension `L1r2` has $w = 1$ in $M$ [computer].
3. **Two carriers** (proposal, not built). Following H: (3.12) shifted to class 5 with a second carrier $x_6$ that repeats the rôle of $x_5$ for the remaining level. Six generators: the dense test at $n = 8$ has $\dim A_8 = 335\,923$ (about 450 GB in `uint32`), beyond any single GPU; even $n = 7$ (55 986) only shows $D_7$. Status: [conjecture], untested.

**(f) Toward** $f_{\mathrm{met}}(5) = 7$ What is proved is about the method, not the groups: for a presentation of the shape (3.12) (paired generators with $x_i^{E_i} \in \gamma_{c-1}$, $w = \prod [x_i, x_j^{A_{ij}}]$), $w - 1$ splits as in Lemma 7; $w \neq 1$ forces a pair with $A_{ij} < E_iE_j$ (Lemma 9); for that pair the pairing lemma certifies the binomial error only up to $\Delta^{c+1}$ (Lemma 10) and the pairing part only up to $\Delta^{k+\min(2+\log_2 m, k)}$. So any proof of $w \in D_8$ at class 5 in this family must get two levels from relators other than the power relators of the paired generators, plus the multiplier 4 (or a cancellation between $S$ and the errors). In every group measured, the levels gained were exactly the carriers' count (Table in (c)). A theorem $f_{\mathrm{met}}(5) = 7$ would need an upper bound for $D_8$ that does not go through this decomposition; nothing here gives it.

**(g) `c5i5` at** $n = 8$**: it fails in** $S$ **and in** $B$ [computer] (Modal A100-SXM4-80GB, main 98e827c, `C4BATCH_APP=math144-c5i5-n8`, `gpu/modal_batch.py spawn --gpu a100 --input results/c5/presentation/records/c5i5_lin.jsonl.gz -- --n 8 --h 1 0 0 0 0 --verbose`; 1 567 s, closure 7 rounds, GPU peak 54.8 GiB.) $M = 97\,655$, 8 054 relator rows, rank 97 648, free columns $7 = r_8$, index 169 278 / 169 264 / 169 194 modulo $2^{32}/2^{30}/2^{20}$ (differences $14 = 2r_8$, $84 = 12r_8$): the certificate holds. Rows in the tables of (b): $w, S \notin \Delta^8$, $XV \in \Delta^8$, $B_{12}, B_{13}, B_{23}, \sum B \notin \Delta^8$, chk $\in \Delta^8$; $128X_2X_1^2 \in \Delta^8$, $128X_2^2X_1 \notin \Delta^8$. Line and log: `results/c5/presentation/c5i5_lin_n8_a100.out.jsonl`, `.log`.

So `c5i5` fails at $D_8$ exactly as c4a fails at $D_7$ and `L2g` at $D_7$: in the pairing part $S$ (multiplier 2 where 4 is needed, (b)) **and** in the binomial error of the $w$-pair, through the same monomial $2E_2X_2^2X_1$. The carrier $x_5$ does not lift the binomial error from the certified $\Delta^{c+1} = \Delta^6$ to $\Delta^8$: at most one level, as at class 4 (the parts were not measured at $n = 7$; there only $w \in D_7$ is known, §6.10). This is what conjecture H predicts; the alternative "$S$ only" (H false) is excluded. H stays a conjecture (two groups, one per class).

**Next.** (1) Done: (g). (2) The identity behind "$x_5$ lifts $128X_2^2X_1$" on paper — that would say what a second carrier must look like (done: §6.12). (3) A test of $D_8$ for six generators needs a method other than the dense form (standard monomials, or a grading).

**Rerun.**

```
# the c4a record (and the c4b record of the table)
scripts/gap5.sh c5_ep.g 'P5NAME := "c4a"; P5I5 := rec(cls := 4);'
python3 scripts/c5_lintests.py IN OUT --A 256 16384 32768
python3 scripts/c5_lintests_extra.py IN OUT 'T12=128*X2*X2*X1'
python3 gpu/howell_gpu.py --device numpy plateau --input OUT.gz --n 6 --h 1 0 0 0 --steps host
# the split and root extensions
mkdir -p gpu/data/plateau5/spl && scripts/gap5.sh c5_split.g 'Read("c5_spl_eq.g"); SPLDIR := "/data/spl";'
scripts/gap5.sh c5_roots.g 'ROOTBASE := "L1"; ROOTM := 2;'
```

Logs and lines: `results/c5/presentation/`; the records with `lintests`: `results/c5/presentation/records/`.

**Modal spend for §6.11:** about $1.4 for (g) (one A100-80GB run, 26 min with the container, app `math144-c5i5-n8`, stopped); everything else local (GAP in Docker, the closures on the CPU).

### 6.12. D5-10: the carrier identity, why it buys one level, and what a class-5 design needs

The question of §6.11(b): which identity in $\mathbb{Z}[G]$ puts $128X_2^2X_1$ into $\Delta^6$ in c4a ((3.12) with $x_6 \to x_3$), and only with the carrier relator $x_5^{1024}d_2^{16}$. Notation as in §6.11: $t = [x_4, x_3]$ (class 4) or $[x_4, x_3, x_3]$ (class 5), $d_l = [t, x_l] \in \gamma_k$, $k = c-1$, $X_i = x_i - 1$, $D_l = d_l - 1$, $E_i = 2^{e_i}$ the order of $x_i$ modulo $\gamma_k$, $P_i = x_i^{E_i} \in \gamma_k$. In c4a: $P_1 = d_2^{-1}d_3^{-4}$, $P_2 = d_1d_3^{16}d_5^{-16}$, $P_3 = d_1^4d_2^{-16}$, $x_5^{1024} = d_2^{-16}$, $E = (8, 64, 16384)$, $(A_{12}, A_{13}, A_{23}) = (256, 16384, 32768)$. Data: `results/c5/carrier/`.

**Summary.**

- [on paper, every step checked by computer] The identity is short. Move the scalar $128 = 2E_2 = 16E_1$ from $X_2$ onto $X_1$:

$$128X_2^2X_1 \equiv X_2^2\,(x_1^{128} - 1), \qquad x_1^{128} = P_1^{16} = d_2^{-16}d_3^{-64} = x_5^{1024}\,d_3^{-64} \pmod{\Delta^6},$$

  then $X_2^2(x_5^{1024} - 1) \equiv 1024X_2^2X_5 \in \Delta^6$, because $512X_2^2 \in \Delta^5$ (Lemma 13, a "quadratic pairing"), and $64X_2^2D_3 = (64X_2)X_2D_3 \in \Delta^{2+1+3}$. Only the relators of $x_1$, $x_2$, $x_5$ and class $\le 4$ are used (Theorem 14). Without the carrier the same steps end in $-16X_2^2D_2$, which is $\notin \Delta^6$ [computer].
- [on paper] So the carrier turns the $d_2$-part $d_2^{-16}$ of $x_1^{128}$ — a *small* power of an element of $\gamma_k$, which $X_2^2$ certifies only to $\Delta^5$ — into a *large* power $x_5^{1024}$ of a generator, which the quadratic pairing absorbs. The $d_3$-part $d_3^{-64}$ is not carried: plain pairing puts it in $\Delta^{k+3}$, which is $\Delta^6$ at class 4.
- [on paper] Lemma 15: once the pairing holds with multiplier 1, $w = [x_2, x_1^{A_{12}}] = \prod_l [x_2, d_l]^{-a_lA_{12}/E_1}$ ($P_1 = \prod_l d_l^{-a_l}$), and every $d_l$-part that pairing alone can certify contributes 1. So $w \ne 1$ needs a part that pairing cannot certify, i.e. a carried part: the carrier is forced, not incidental.
- (b) [on paper] One level only: the uncarried part $64X_2^2D_3$ sits exactly at $\Delta^{k+3}$ ($= \Delta^6$ in c4a); everything else of the identity is certified one level higher. The prediction "at $n = 7$ the piece $64X_2^2D_3$ fails and the carrier piece and the binomials of $x_1$ do not" is confirmed [computer, Modal A10G, certified]: the identity holds modulo $\Delta^7$ with only $64X_2^2D_3 \notin \Delta^7$ (see (b)).
- (c) [on paper] Conditions for $NX_2^2X_1 \in \Delta^{2k}$ ($N = A_{12}/2$): each $d_l$-part of $x_1^N$ is either carried by a power $x_\kappa^F$ of a generator with $2^kE_2 \mid F$, or has $v_2(a_lN/E_1) \ge e_2 + k - 3$ (Proposition 16, sufficient); a second carrier is called for when two parts fail the second condition. For `c5i5` the $d_3$-part fails ($64 < 2E_2$), which is why $128X_2^2X_1 \notin \Delta^8$ there (§6.11(g)). [computer] Seven presentations built: the second carrier on `c5i5` (`c5i6`, six generators), two multiplier-4 designs and four one-change variants — $w = 1$ **in** $M = \Gamma/\gamma_6\Gamma''$ **in every one** ((d)). The only group of this family with $w \ne 1$ at class 5 is still `c5i5`.

**(a) The identity** **Which relators carry it** [computer, `scripts/c5_carrier_ablate.sh`, $n = 6$, CPU]. With the test record of §6.11 (c4a = $G$ of `plateau5_c4i5`, 1 554 relators: the 9 of (3.12), 45 metabelian, 250 of $\ker\varphi$, 1 250 commutators of weight 5), dropping the 5 relators $d_l^{N_l}$ and the two top ones, or the 45 metabelian ones, or the 250 of $\ker\varphi$, keeps $128X_2^2X_1 \in \Delta^6$ (certified); dropping the relator of $x_3$ keeps it too (membership found, the certificate needs other $h$ and was not redone); dropping the relator of $x_1$ or of $x_2$ loses it (not certified for the same reason), and so does dropping that of $x_5$ (§6.11, certified). Moving the scalar around: in c4a, c4b and c4a without $x_5$'s relator ($h = (2, 1, 2, 0)$, $r_6 = 47$, certificate holds), with $e_1 = 2D_1X_2X_1$, $e_2 = 32D_3X_2X_1$, $e_3 = 32D_5X_2X_1$, $e_4 = 32X_2^4X_1$: $128X_2^2X_1 \equiv e_1 + e_2 - e_3 + 3969\,e_4$ (the relator of $x_2$), $e_2, e_3, e_4 \in \Delta^6$ in all three, and $e_1 \in \Delta^6$ only in c4a: the whole effect sits in $2D_1X_2X_1$, which is again $128X_2^2X_1$ read through the relator of $x_2$. The way out is to read the scalar through the relator of $x_1$ instead.

**Lemma 13** [on paper] (quadratic pairing). Let $k \in \{3, 4\}$, $G$ of class $\le k+1$, $x^E = P = \prod_l d_l^{e_l}$ with $d_l \in \gamma_k$, $E = 2^e \ge 4$, $X = x - 1$, $s \in \{0, 1\}$, $2 + s \le k$. Suppose that for every $l$ either $E \mid 8e_l$, or $d_l^{2^{s+3}e_l} \in \gamma_{k+1+s}$. Then $2^{s+3}EX^2 \in \Delta^{k+2+s}$.

*Proof.* $x^E - 1 = \sum_{m\ge1}\binom{E}{m}X^m$, so $2^{s+3}EX^2 = 2^{s+3}(P-1)X - \sum_{m\ge2}2^{s+3}\binom{E}{m}X^{m+1}$. Modulo $\Delta^{2k}$, $P - 1 \equiv \sum_l e_l D_l$. A term $2^{s+3}e_lD_lX$ is $D_l \cdot \frac{8e_l}{E}(2^sEX) \in \Delta^{k+2+s}$ in the first case (pairing lemma of §6.7: $2^sEX \in \Delta^{2+s}$), and $(d_l^{2^{s+3}e_l} - 1)X \in \Delta^{k+2+s}$ up to $\Delta^{2k}$ in the second. For $m = 2$: $2^{s+2}E(E-1)X^3$ is $(4EX)X^2 \in \Delta^{\min(4,k)+2} = \Delta^{k+2}$ for $s = 0$ and $(E-1)X(8EX^2) \in \Delta^{k+3}$ for $s = 1$ (by $s = 0$). For $m \ge 3$, $v_2\binom{E}{m} = e - v_2(m)$, so the term is a unit times $(2^{s+3-v_2(m)}EX)X^m$, of level $\ge \min(2+\max(0, s+3-v_2(m)), k)+m \ge k+2+s$. □

Where the second case comes from [on paper]: if $t \in \gamma_{k-1}$, $d_1 = [t, x_1]$ and $x_1^{E_1} \in \gamma_k$, then $1 = [t, x_1^{E_1}] = d_1^{E_1}[d_1, x_1]^{\binom{E_1}{2}}$ (class $k+1$, $[\gamma_{k-1}, \gamma_k] \subseteq \gamma_{2k-1} = 1$), so $d_1^{E_1} \in \gamma_{k+1}$, and $[d_1, x_1]^{E_1} = [d_1^{E_1}, x_1] = 1$ gives $d_1^{2E_1} = 1$. For c4a ($x = x_2$, $E = 64$, $e_1 = 1$, $e_3 = 16$, $e_5 = -16$, $E_1 = 8$): $512X_2^2 \in \Delta^5$ and $1024X_2^2 \in \Delta^6$. Measured [computer]: $2^sX_2^2 \in \Delta^5$ iff $s \ge 9$, and $512X_2^2 \in \Delta^6$ already (better than certified); $2^sX_2 \notin \Delta^4$ for all $s \le 14$ — the gain is genuinely quadratic, not a better linear pairing.

An explicit form at $n = 5$ [computer, `scripts/c5_carrier_rows.py --ddmin`, `scripts/c5_carrier_solve.sage`]: 14 shifted relator rows suffice (only $r_1 = x_1^8d_2d_3^4$ and $r_2 = x_2^{64}d_1^{-1}d_3^{-16}d_5^{16}$, no commutator relator), and over $\mathbb{Z}$

$$512X_2^2 \equiv 8X_2(r_2-1) - 252X_2^2(r_2-1) + 2730X_2(r_2-1)X_2^2 - 2(r_2-1)\big(\Lambda_3+\Lambda_5\big) + X_2\big[(r_1-1),\, X_3X_4 - X_4X_3\big] \pmod{\Delta^5},$$

$\Lambda_3 = -X_3X_3X_4 + 2X_3X_4X_3 - X_4X_3X_3$, $\Lambda_5 = -X_3X_4X_5 + X_4X_3X_5 + X_5X_3X_4 - X_5X_4X_3$ (the leading forms of $D_3$, $D_5$ up to sign), $[a,b] = ab - ba$: the three ingredients of the proof — the relator of $x_2$ with its binomial corrections, $2D_l \cdot 64X_2$, and $[t, x_1^8] = 1$ — appear literally (this is the mirror image, $X_2$ on the left).

**Theorem 14** [on paper]. Let $G$ have class $\le 4$, $t \in \gamma_2$, $d_l = [t, x_l]$, and $x_1^8 = d_2^{-1}d_3^{-4}$, $x_2^{64} = d_1d_3^{16}d_5^{-16}$, $x_5^{1024} = d_2^{-16}$. Then $128X_2^2X_1 \in \Delta^6$.

*Proof.* (1) $128X_2^2X_1 = X_2^2(x_1^{128} - 1) - \sum_{m=2}^{5}\binom{128}{m}X_2^2X_1^m$. The sum is in $\Delta^6$: $\binom{128}{2} = 127 \cdot 64$ and $64X_1^2 = (8X_1)^2 \in \Delta^4$; $\binom{128}{3} = 2^7 \cdot 2667$ and $128X_1^3 = (8X_1)(16X_1)X_1 \in \Delta^6$; $m \ge 4$ has degree $\ge 6$. (2) $\gamma_3$ is abelian ($[\gamma_3, \gamma_3] \subseteq \gamma_6$), so $x_1^{128} = (d_3^{-4}d_2^{-1})^{16} = d_3^{-64}d_2^{-16} = d_3^{-64}x_5^{1024}$, and $x_1^{128} - 1 = (d_3^{-64} - 1) + (x_5^{1024} - 1) + (d_3^{-64} - 1)(x_5^{1024} - 1)$, the last term in $\Delta^3\Delta^3$. (3) $X_2^2(d_3^{-64} - 1) \equiv -64X_2^2D_3 = -(64X_2)X_2D_3 \in \Delta^{2+1+3}$. (4) $X_2^2(x_5^{1024} - 1) = \sum_{m\ge1}\binom{1024}{m}X_2^2X_5^m$: $m = 1$ is $2(512X_2^2)X_5 \in \Delta^6$ by Lemma 13; $\binom{1024}{2} = 512 \cdot 1023$ gives $(512X_2^2)X_5^2 \in \Delta^7$; $\binom{1024}{3}$ is divisible by 1024; $m \ge 4$ has degree $\ge 6$. □

So, as an identity:

$$128X_2^2X_1 \;\equiv\; \underbrace{X_2^2(x_5^{1024}-1)}_{\text{carrier, Lemma 13}} \;-\; \underbrace{64X_2^2D_3}_{\text{pairing}} \;-\; \underbrace{\textstyle\sum_{m\ge2}\binom{128}{m}X_2^2X_1^m}_{\text{binomials of } x_1} \pmod{\Delta^6}.$$

The relators used: of $x_1$ (16 times, through $x_1^{128} = P_1^{16}$), of $x_5$ (once), of $x_2$ (inside Lemma 13), and class $\le 4$; not the relator of $x_3$, not $G'' = 1$, not $d_l^{N_l}$ or the top relators — as the ablation says. Without the carrier, (2) ends in $d_2^{-16}$ and (4) in $-16X_2^2D_2 = -(16X_2)X_2D_2$, certified only in $\Delta^5$; measured $\notin \Delta^6$ in c4b and in c4a without $x_5$'s relator [computer].

[computer, `results/c5/carrier/t6.py`, c4a, $n = 6$, certificate holds] every piece: $128X_2^2X_1 - X_2^2(x_1^{128} - 1)$, $x_1^{128} - x_5^{1024} + 64D_3$, $X_2^2(x_5^{1024} - 1)$, $64X_2^2D_3$, $1024X_2^2X_5$, the binomials of $x_1$ and of $x_5$, $512X_2^2$, $8D_1X_2$, $128D_3X_2$, $128D_5X_2$, and $128X_2^2X_1 - \big(X_2^2(x_5^{1024} - 1) - 64X_2^2D_3 - \sum_{m\ge2}\binom{128}{m}X_2^2X_1^m\big)$ are all in $\Delta^6$.

**Lemma 15** [on paper] (where $w$ lives). Let $G$ have class $c = k+1$, $x_i^{E_i} = P_i \in \gamma_k$, $E_j \mid A_{ij}$, and $g_1 \in (\gamma_k)^{E_1}\gamma_c$, $g_2 \in (\gamma_k)^{E_2}\gamma_c$ (the pairing with multiplier 1). Then

$$w = [x_2, x_1^{A_{12}}] = \prod_l [x_2, d_l]^{-a_lA_{12}/E_1} \qquad (P_1 = \textstyle\prod_l d_l^{-a_l}).$$

*Proof.* For $y, z \in \gamma_k$, $[x_i, yz] = [x_i, y][x_i, z]$ and $[x_i, y^{E_i}] = [x_i^{E_i}, y] = [P_i, y] = 1$ ($[x_i, y]$ is central, $\gamma_k$ is abelian). So $u_{12}u_{13} = [x_1, g_1] = 1$ and $u_{23} = [x_2, h_{23}] = [x_2, g_2][x_2, h_{21}]^{-1} = [x_2, x_1^{A_{12}}]$. □

Since $[x_2, d_l]^{E_2} = 1$, a $d_l$-part with $E_2 \mid a_lA_{12}/E_1$ contributes nothing. In c4a and `c5i5` the pairing holds with multiplier 2 (§6.11(b)), so $w = [x_2, d_2]^{-32}[x_2, d_3]^{-128} = [x_2, d_2]^{-32}$: $w$ **is exactly the $d_2$-part, the carried one.** This is not a coincidence (Proposition 16 below): a part that pairing alone can certify has $E_2 \mid a_lA_{12}/E_1$ and dies in $w$; so $w \ne 1$ needs a part that must be carried.

**(b) Why one level** Run the proof of Theorem 14 against $\Delta^7$ (c4a, $k = 3$). Steps (1) and (4) go one level further: $64X_1^2 = 8E_1X_1^2 \in \Delta^5$ (Lemma 13 for $x_1$, $s = 0$: $e_2 = 1$, $E_1 \mid 8$; $e_3 = 4$) gives the binomials of $x_1$ in $\Delta^7$, and $1024X_2^2X_5 = (16E_2X_2^2)X_5$ with $1024X_2^2 \in \Delta^6$ (Lemma 13, $s = 1$) is in $\Delta^7$. Step (3) does not: $64X_2^2D_3 = (E_2X_2)X_2D_3$ has the certified level $2 + 1 + 3 = 6$ and no more, and 64 is the exponent that the relator of $x_1$ gives ($a_3N/E_1 = 4 \cdot 128/8$). So the carrier lifts its own part (the $d_2$-part) by two levels over the pairing (from $1 + 1 + k$ to $\ge 2k + 1$), and the stop is in the part it does not carry. [computer] (Modal A10G, ref `ae122f7`, app `math170-n7`, `gpu/modal_batch.py spawn --ref ae122f7 --gpu a10g`

`--input results/c5/carrier/records/c4a_pieces.jsonl.gz -- --n 7 --h 1 0 0 0 --verbose`; 91 s, closure 6 rounds, GPU peak 5.5 GiB; the local CPU attempt `c4a_pieces_n7.log` had been stopped after round 2.) $M = 19\,530$, 1 554 relator rows, rank 19 524, free columns $6 = r_7$, index 34 830 / 34 818 / 34 758 modulo $2^{32}/2^{30}/2^{20}$ (differences $12 = 2r_7$, $72 = 12r_7$): **the certificate holds**; rank, index and the five $w$-tests coincide with `c4a_lin2_n7.log`. Membership in $\Delta^7$ (c4a):

| piece | $\in \Delta^7$ |
|---|---|
| $128X_2^2X_1$ ($T_{12}$) | no |
| $X_2^2(x_5^{1024}-1)$ (carried $d_2$-part) | **yes** |
| $64X_2^2D_3$ (uncarried $d_3$-part) | **no** |
| $\sum_{m\geq 2}\binom{128}{m}X_2^2X_1^m$ (binomials of $x_1$) $= -(128X_2^2X_1 - X_2^2(x_1^{128}-1))$ | **yes** (both forms) |
| $8128X_2^2X_1^2$ (its $m=2$ term) | yes |
| $1024X_2^2X_5$; binomials of $x_5$ | yes; yes |
| $128X_2^2X_1 - \left(X_2^2(x_5^{1024}-1) - 64X_2^2D_3 - \sum_{m\geq 2}\binom{128}{m}X_2^2X_1^m\right)$ (the identity's remainder) | **yes** |
| $x_1^{128} - x_5^{1024} + 64D_3$ (no $X_2^2$ factor; enters only as $X_2^2\cdot(\ldots)\in\Delta^8$) | no |
| $512X_2^2$, $1024X_2^2$, $2048X_2^2$ | no, no, no |
| $8D_1X_2$; $128D_3X_2$, $128D_5X_2$ | yes; no, no |
| $16X_2^2D_2$ (the $d_2$-part read without the carrier) | yes |

So the identity of Theorem 14 holds modulo $\Delta^7$ with every term in $\Delta^7$ except $64X_2^2D_3$: $128X_2^2X_1 \equiv -64X_2^2D_3 \pmod{\Delta^7}$, and **the uncarried $d_3$-part is the only obstruction at $\Delta^7$ in c4a** — the prediction of this subsection is confirmed [computer]. $1024X_2^2 \in \Delta^6$ ($n=6$) but $2^sX_2^2 \notin \Delta^7$ for $s \leq 11$: the carrier term is in $\Delta^7$ through $(1024X_2^2)X_5 \in \Delta^6\Delta$, as in the proof. A side observation: $16X_2^2D_2 \in \Delta^7$ in c4a (it was $\notin \Delta^6$ in c4b and without $x_5$'s relator), i.e. once $x_5$'s relator is present the $d_2$-part is certified also when read directly; the proof of that through Lemma 13 is not written [conjecture: it is $(16X_2)X_2D_2$ with $d_2^{16} = x_5^{-1024}$ and Lemma 13 for $x_5$]. Line and log: `results/c5/carrier/c4a_pieces_n7_a10g.out.jsonl`, `.out.log`.

**(c) The conditions, and a second carrier** **Proposition 16** [on paper]. Let $k \in \{3,4\}$, $G$ of class $k+1$, $P_1 = \prod_l d_l^{-a_l}$, $N = A_{12}/2 = 2^\nu$. Suppose (i) Lemma 13 holds for $x_2$ with $s = k-3$; (ii) $\frac{N}{2}X_1^2 \in \Delta^{2k-2}$ (for $k=3$: $N \geq 2E_1^2$; for $k=4$: Lemma 13 for $x_1$ with $s=0$ and $N \geq 16E_1$); (iii) for every $l$ either

- (pairing) $v_2(a_lN/E_1) \geq e_2 + k - 3$, or
- (carrier) $d_l^{-a_lN/E_1} = x_\kappa^F$ for a generator $x_\kappa$ and $2^kE_2 \mid F$.

Then $NX_2^2X_1 \in \Delta^{2k}$. *Proof:* Theorem 14 with these numbers: an uncarried part is $(\frac{a_lN}{E_1}X_2)X_2D_l$ with $2^{e_2+k-3}X_2 \in \Delta^{k-1}$; a carried one is $X_2^2(x_\kappa^F - 1)$, whose first term is $(2^kE_2X_2^2)X_\kappa$ with $2^kE_2X_2^2 \in \Delta^{2k-1}$ (Lemma 13 with $s=k-3$), and whose second has $v_2\binom{F}{2} \geq e_2 + k - 1$. □

And by Lemma 15 the pairing-type parts are invisible in $w$ (for $k \geq 3$, $v_2(a_lA_{12}/E_1) \geq e_2 + k - 2 \geq e_2$), so the $w$-carrying part of $P_1$ must satisfy the carrier condition. **So Proposition 16 calls for a second carrier when two $d_l$-parts fail the pairing condition.** The data:

| group | $k$ | $d_2$-part of $x_1^N$ | $d_3$-part | $NX_2^2X_1$ (predicted) | measured |
|---|---|---|---|---|---|
| c4b | 3 | $d_2^{-16}$, not carried, $v_2 = 4 < 6$ | $d_3^{-64}$, $6 \geq 6$ | not certified at $\Delta^6$ | $\notin \Delta^6$ |
| c4a | 3 | carried, $x_5^{1024}$, $2^3E_2 = 512 \mid 1024$ | $6 \geq 6$ | $\in \Delta^6$ | $\in \Delta^6$ |
| c4a, target $\Delta^7$ | 3 | carried (Lemma 13, $s=1$) | $6 < 7$ | not certified at $\Delta^7$ | $\notin \Delta^7$ (§6.11); only $64X_2^2D_3 \notin \Delta^7$ ((b)) |
| `c5i5` | 4 | carried, $2^4E_2 = 1024 \mid 1024$ | $6 < 7$ | not certified at $\Delta^8$ | $\notin \Delta^8$ (§6.11(g)) |
| `c5i6` (below) | 4 | carried by $x_5$ | carried by $x_6^{1024}$ | $\in \Delta^8$ | — ($d=6$, $n=8$ out of reach) |

So conjecture H of §6.11 should be read as: a carrier repairs one $d_l$-part completely; how many levels that shows in $w$ depends on the worst *uncarried* part, which sits at $\Delta^{k+3} = \Delta^{c+2}$ in (3.12) and `c5i5`. That is the observed "one carrier, one level".

**(d) Class-5 candidates built in GAP** [computer, `scripts/c5_carrier_gap.sh` = `gap/c5_ep.g` on `I5cand` records, Docker, local; logs `results/c5/carrier/ep_*.log`]

| name | $d$ | change against `c5i5` | multiplier (by design) | Prop. 16 for $128X_2^2X_1$ at $\Delta^8$ | $\lvert w\rvert$ i |
|---|---|---|---|---|---|
| `c5i5` (§6.7) | 5 | — | 2 | $d_3$-part fails | 2 |
| `c5i6` | 6 | $+x_6$, relator $x_6^{1024}d_3^{64}$ (second carrier) | 2 | holds | **1** |
| `c5m4` | 5 | $P_1 = d_2^{-4}d_3^{-8}$, $P_3 = d_1^{-4}d_2^{-32}$, $x_5^{1024} = d_2^{-64}$, $A_{23} = 4E_3$ | 4 | holds | **1** |
| `c5m4b` | 5 | $P_1 = d_2^{-1}d_3^{-8}$, $P_2 = d_1^4d_3^{64}d_5^{-16}$, $P_3 = d_1^8d_2^{-16}$, $A_{13} = A_{23} = 2E_3$ | 4 | holds | **1** |
| `v1` | 5 | $A_{13} = 2E_3$ only | — | fails | **1** |
| `v2` | 5 | $P_1 = d_2^{-1}d_3^{-8}$ only | — | holds | **1** |
| `v3` | 5 | $P_2 = d_1^4d_3^{64}d_5^{-16}$ only | — | fails | **1** |
| `v4` | 5 | $P_3 = d_1^8d_2^{-16}$ only | — | fails | **1** |

(All other relators as in `c5i5`: $d_1^{16}$, $d_2^{128}$, $d_5^{2048}$, the two top relators. Multipliers by design, from the exact $g_i$ in $\gamma_4$: for `c5m4b`, $g_1 = d_1^{32}d_2^{-32}d_3^{256}d_5^{-64} \in (\gamma_4)^{32}$, $g_2 = d_1^{16}d_3^{256} \in (\gamma_4)^{256}$, $g_3 = 1$; `c5_ep.g` stops at "$w = 1$ in $M$" before its own multiplier report. Each run 1–2 h of GAP/nq in Docker on the loaded machine. `c5m4x6` = `c5m4` $+x_6^{1024}d_3^{128}$ and `c5m4b6` = `c5m4b` $+x_6^{1024}d_3^{128}$ were stopped by hand: a presentation that contains all relators of another one, on more generators, maps $w \mapsto w$, so $w = 1$ there too [on paper].)

What the table says:

- `c5m4` [on paper, Lemma 15]: with multiplier $\geq 1$, $w = [x_2, P_1^{32}] = [x_2, d_2]^{-128}[x_2, d_3]^{-256}$ and $[x_2, d_l]^{64} = 1$: $w = 1$ is forced. Raising $a_2$ to get the pairing kills the $w$-part — Lemma 15 in action.
- `c5m4b` keeps $a_2 = 1$, so Lemma 15 gives $w = [x_2, d_2]^{-32}$, as in `c5i5`; still $w = 1$: here the order of $[x_2, d_2]$ drops. The one-change variants show that **each** of the four changes does this alone; and so does the second carrier `c5i6`, which touches neither $P_1$, $P_2$, $P_3$ nor $A$. [conjecture] The order of $[x_2, d_2]$ in $\gamma_5$ is tied to the $d_3$-part through the relator of $x_1$ (in the Lie ring, $[x_1^8 d_2 d_3^{a_3}, x_2]$ relates $[d_2, x_2]$ to $a_3[d_3, x_2]$ and to $8[x_1, x_2]$), so making the $d_3$-part harmless for $T_{12}$ — by carrying it or by $a_3 = 8$ — also makes $[x_2, d_2]^{32}$ trivial. Not proved.
- Dense tests: no candidate with $w \neq 1$, so none was run ($\dim A_6 = 9\,330$ for $d = 6$ would be fine on the CPU, $\dim A_7 = 55\,986$, $\dim A_8 = 335\,923$ not).

**(e) Status**

- Proved [on paper]: Lemma 13, Theorem 14 (the identity for c4a), Lemma 15, Proposition 16; the exact $n = 5$ form of Lemma 13 [computer, over $\mathbb{Z}$].
- Computed [computer, certified]: every piece of Theorem 14 in $\Delta^6$ (c4a), $e_1 \notin \Delta^6$ in c4b and without the carrier relator, the ablation, the levels of $2^sX_2$ and $2^sX_2^2$; GAP: $w = 1$ in all seven new presentations. At $n = 7$ (A10G): every piece except $64X_2^2D_3$ in $\Delta^7$, so the $d_3$-part is the only obstruction at $\Delta^7$ in c4a ((b)).
- Conjectured: that in the family (3.12) with $t = [x_4, x_3, x_3]$ the $T_{12}$-condition of Proposition 16 at class 5 and $w \neq 1$ exclude each other ((d)).
- Not done: a class-5 presentation with multiplier 4, both $T_{12}$-parts repaired and $w \neq 1$. The next candidates should keep $P_1 = d_2^{-1}d_3^{-4}$ and $A$ of `c5i5` and repair the $d_3$-part through a generator that does not enter the Lie relation of $x_1$ with $x_2$ (e.g. a carrier for $d_3^{64}$ placed in the relator of $x_3$ instead of a new power relator) — untested.

**Rerun.**

```
# tests (about 15 min on the loaded machine)
python3 scripts/c5_carrier_tests.py results/c5/presentation/records/c4a_lin2.jsonl.gz OUT.jsonl.gz --replace
python3 gpu/howell_gpu.py --device numpy plateau --input OUT.jsonl.gz --n 6 --h 1 0 0 0 --steps host
python3 results/c5/carrier/summ.py LOG    # prints a table
# ablation
scripts/c5_carrier_ablate.sh BASE TESTS.py 6 TAG "4-8" "54-303" "0" ...
# minimal rows and the exact identity
python3 scripts/c5_carrier_rows.py results/c5/presentation/records/c4a_lin2.jsonl.gz --n 5 --target '512*X2*X
python3 scripts/c5_carrier_exact.py OUT.json EX.json
sage scripts/c5_carrier_solve.sage EX.json
# GAP
scripts/c5_carrier_gap.sh c5i6 c5m4 c5m4b v1 v2 v3 v4
```

**Modal spend for §6.12:** $0.05 (one A10G call, 91 s of compute, app `math170-n7`, stopped; `modal billing report`: $0.054); everything else local (CPU closures, GAP in Docker, Sage).

### 6.13. D5-11: $J_n$ without the dense form — standard monomials, level by level

The question: test $w-1 \in J_n$ for $d=6$ generators at $n=7,8$, where the dense form of §6.5 has $M = 55\,986$ and $335\,922$ columns ($\approx 450$ GB in `uint32` at $n=8$). Here $J_n$ is built degree by degree in the columns of *candidate* monomials only; on every group we have it gives the same index, generators and memberships as the dense closure, and the full reduced Howell form rebuilt from it is byte for byte the dense one. Code: `gpu/stdmono.py` (with `gpu/sparse_howell.c` for `--sparse`, `gpu/modal_stdmono.py` for Modal). Data: `results/nondense/`.

**Summary.**

- [on paper] $J_k = \mathrm{span}\{\mu(r)-1\} + J_{k-1}a + aJ_{k-1}$ (Lemma N1): one Howell form per level, no closure rounds. A word whose prefix or suffix is not a standard monomial of the previous level is rewritten by a fixed rule into later columns (a projection $\pi_k$ with kernel in $J_k$), so only the candidates $C_k$ stay as columns; the generators of $\pi_k(J_k)$ are the relators, both shifts of the torsion rows, and few "overlaps" (Lemma N2).
- [computer] Validation: c4a $n = 5,6,7$, `c5i5` $n = 6,7,8$, the $M$ of §6.10 (q5) $n = 6,7,8$: index modulo $2^{32}/2^{30}/2^{20}$, generators, all tests and lintests as in the dense runs; SHA-256 of the full reduced form equal to the dense one at c4a $n = 5,6,7$ and `c5i5` $n = 6,7$. `c5i5` at $n=8$ reproduces §6.11(g) on an A10G in 5.4 min (dense: 26 min on an A100), with $45\,530$ columns instead of $97\,655$.
- [computer] The reduced form is very sparse in these columns: 0.57 M nonzeros in $45\,530$ rows (`c5i5`, $n=8$), and the generator rows too (14 nonzeros per torsion shift). `--sparse` (row by row, in C) uses this.
- [computer] `--sparse` at $n=8$ (`c5i5`, A10G): the §6.11(g) answers again, in 126 s of computation (155 s with the container), GPU peak 2.0 GiB, host peak 8.7 GiB; the form has $573\,467$ nonzeros (canonical: $565\,950$) — no fill-in.
- [estimate] $d=6$: $n=7$ is easy (about 34–41 thousand columns; minutes on one GPU, dense or sparse). $n=8$: 154–203 thousand columns; the dense form (48–82 GB) does not fit one 80 GB GPU, but `--sparse` does: about 25–35 min on one A10G (15–20 min on an A100), GPU memory $\approx$ 8–12 GB, host memory $\approx$ 20–25 GB. Details in (d).

**(a) Structure and the options, with sizes** Columns in increasing degree, as everywhere in B. Facts used [on paper, §3.4]: the Howell form of $J_n$ truncated to degrees $< k$ is a Howell form of $J_k$; the columns without a unit pivot (the *standard monomials* $S_k$) are closed under prefixes and suffixes (if $u$ has a unit pivot, $u - \mathrm{NF}(u) \in J$ with $\mathrm{NF}(u)$ on later columns, and multiplying by $a_j$ on either side keeps "later"); $\mathrm{NF}_k(u)$ lies in the standard monomials of degree $\geq \deg u$. The number of standard monomials is at least $\dim_{\mathbb{F}_2} \Delta_{\mathbb{F}_2}/\Delta^n_{\mathbb{F}_2}$ (`generators`, Jennings, §6.4), a lower bound for every monomial basis.

| option | columns, `c5i5` $n=8$ (measured) | $d=6$, $n=8$ (model of (d)) |
|---|---|---|
| dense (§6.5) | $97\,655$ | $335\,922$ |
| levels in all columns (Lemma N1 only) | $97\,655$ | $335\,922$ |
| only the top degree compressed | $19\,530 + 33\,709$ | $\approx 56\,000 + 130\,000$ |
| candidates $C_n$ (chosen) | $45\,530$ | $154\,000$–$203\,000$ |
| lower bound: generators | $44\,370$ | $\geq$ about $150\,000$ |
| candidates, sparse form | $45\,530$ | the same |
| the subspace reachable from $w-1$, or the dual | — | no gain: membership needs the whole lattice bel |
| the metabelian relators $\Gamma''$ | — | already used: they are relators; their effect *is* the |

A "top-down" order of the degrees does not help either: the generators are a property of the quotient, not of the order. So the chosen method is the candidate columns, level by level, dense or sparse.

**(b) The method** Notation: $A_k$ has the degrees $< k$; $J_{k-1} \subseteq A_{k-1}$ is the truncation of $J_k$ (§3.4); $S_k$, $\mathrm{NF}_k$ as above.

**Lemma N1** [on paper]. $J_k = \mathrm{span}\{\mu(r)-1 \,:\, r \in R_0\} + \sum_j \left(J_{k-1}a_j + a_jJ_{k-1}\right)$, the products taken in $A_k$ (also modulo $2^E$). *Proof.* For $z \in J_{k-1}$ take a lift $\tilde{z} \in J_k$; $\tilde{z}a_j = za_j$ in $A_k$ because the degree-$(k-1)$ part of $\tilde{z}$ dies in the product; so "$\supseteq$". Conversely $J_k$ is spanned by $u(\mu(r)-1)v$ with monomials $u, v$; if $v = v'a_j$, then $u(\mu(r)-1)v = \left(u(\mu(r)-1)v'\right)a_j$ and the bracket is in $J_k$, its truncation in $J_{k-1}$; the same on the left. □

**Candidates and the projection.** $C_k$: the words of degree 1 and the words $v$ of degree $p \geq 2$ with $v[:-1], v[1:] \in S_{k-1}$. Since $S_k \cap (\deg < k-1) = S_{k-1}$ and $S_k$ is closed under prefixes and suffixes, $S_k \subseteq C_k$. For $v \notin C_k$: if $u = v[:-1] \notin S_{k-1}$,

$v \mapsto \mathrm{NF}_{k-1}(u)\,a_l$ (prefix rule); otherwise $v = a_f u'$ with $u' \notin S_{k-1}$, $v \mapsto a_f\,\mathrm{NF}_{k-1}(u')$ (suffix rule). Each step is a congruence modulo $J_k$ ($u - \mathrm{NF}_{k-1}(u)$ is the truncation of an element of $J_k$; the difference is in degree $k-1$ and dies after the shift) and moves to strictly later columns; $\pi_k$ is the result. So $x - \pi_k(x) \in J_k$, $\pi_k$ is the identity on $C_k$, and

$$x \in J_k \iff \pi_k(x) \in \pi_k(J_k), \qquad A_k^+/J_k \cong \mathbb{Z}^{C_k}/\pi_k(J_k)$$

— the index, the free columns and `generators` are those of the dense form, so the certificate of §6.4 applies unchanged.

**Lemma N2** [on paper]. $\pi_k(J_k)$ is generated by $\pi_k$ of: (i) $\mu(r) - 1$; (ii) $ta_j$, $a_j t$ for the rows $t$ of the Howell form of $\pi_{k-1}(J_{k-1})$ with a non-unit pivot; (iii) the *overlaps* $a_j\big(u - \mathrm{NF}_{k-1}(u)\big)$ for $u \notin S_{k-1}$ with $u[:-1] \in S_{k-1}$ and $a_j\,u[:-1] \notin S_{k-1}$. *Proof.* By N1, with the generating set $\{g_u = u - \mathrm{NF}_{k-1}(u) : u \notin S_{k-1}\} \cup \{\text{torsion rows}\}$ of $J_{k-1}$ (the reduced Howell form in all columns). $\pi_k(g_u a_j) = 0$ by the prefix rule. $\pi_k(a_j g_u) = 0$ when $a_j u[:-1] \in S_{k-1}$ (the suffix rule applies to $a_j u$). If $u = u' a_l$ with $u' \notin S_{k-1}$: $z = g_u - g_{u'} a_l \in J_{k-1}$ lies on columns after $u$, and $a_j g_u = (a_j g_{u'}) a_l + a_j z$ in $A_k$; the first term is a right shift of an element of $J_{k-1}$ (its $\pi_k$ is in the span of (ii)), and $z$ is a combination of $g_x$, $x > u$, and torsion rows (Howell property); induction downwards on $u$. □

Then $H_k$ is the Howell form of these rows in the columns $C_k$, and $\mathrm{NF}_k(u)$ for $u \notin S_k$ is $\pi_k(u)$ reduced by $H_k$, kept as matrices $T_k[p]$ (rows: non-standard words of degree $p$, columns: candidates of degree $\geq p$) for the prefix and suffix rules of level $k+1$. The truncation property is checked at every level (pivots and valuations below the top degree equal those of $H_{k-1}$).

**Implementation** [computer]. `gpu/stdmono.py run` (same output fields as `howell_gpu.py batch`, plus `columns`, per-level statistics): relators and tests are built once in $A_n$ (binomial expansion for powers of long words: $n-2$ products instead of binary powering) and truncated per level; $\pi_k$ is two GEMMs per letter with $T_{k-1}$, repeated until no non-candidate column is left; the Howell form and its images modulo $2^{30}, 2^{20}, 2$ by `howell_gpu.py` (numpy or cupy), or with `--sparse` by `gpu/sparse_howell.c`: rows inserted one by one, the latest leading column first, each new pivot row reduced by the pivots already there (in generation order and unreduced the fill-in was 12 times larger: c4a $n = 6$, 210 497 nonzeros against 17 162; $n = 7$, 16.9 M and 23 min in the C insertion, stopped). `--digest` rebuilds the reduced form in all $M$ columns (rows of $H_n$, and $u$ + reduce$(-\pi_n(u))$ for the other words) and prints `howell_gpu.digest`. 400 random matrices: the C form and `howell_gpu.howell` agree in pivots, index, images and membership.

**(c) Validation against the dense closure [computer, certified]** (D5-21, 01.11.2026). The answers of c4a $n = 5, 6, 7$ and `c5i5` $n = 6, 7, 8$ below (every test and lintest, and the index modulo $2^{32}$) are confirmed by `cert/check_in.c`, which shares no code with this method: its own level-by-level sparse closure in all $M$ columns, closure and Howell property checked, exactness checked against $r_n$ (`results/cert/{c4a,c5i5}_n*_E32.claim`, written from these runs by `cert/make_claims.py`). `c5i5` at $n = 8$: 97 648 rows, 4.9 M nonzeros, 32 s and 190 MB on one core of kimmy. Trusted: the presentations; rank $\geq r_n$ (Quillen, $h$ from nq); and for a membership that the element (for a lintest, its image in $\Delta/\Delta^n$) has 2-power order. The last holds because $\gamma_2$ is a finite 2-group ($h_k = 0$ for $k \geq 2$; nq): the tests lie in $\gamma_c$, and the lintests are built from $x_1, x_2, x_3$ only, with $x_i^{E_i} \in \gamma_k$ (§6.12), so every $g_k$ has 2-power order (`cert/README.md`).

[computer] All runs: certificate holds (free columns = $r_n$, index differences $2r_n$, $12r_n$). "Dense" = the earlier runs of §6.10, §6.11, or a run made here (`c5i5` $n = 6$ on the local CPU, $n = 7$ on an A10G, `results/nondense/a10g_n7/c5i5_dense_n7.*`).

| group | $n$ | $M$ | columns | index $2^{32}/2^{30}/2^{20}$ | generators | tests, lintests | SHA-25 |
|---|---|---|---|---|---|---|---|
| c4a | 5 | 780 | 780 | 3 311 / 3 303 / 3 263 | 619 | = | `f1a381` |
| c4a | 6 | 3 905 | 2 626 | 10 834 / 10 824 / 10 774 | 2 325 | = | `56f3da` |
| c4a | 7 | 19 530 | 8 851 | 34 830 / 34 818 / 34 758 | 8 276 | = | `0d3edc` |
| `c5i5` | 6 | 3 905 | 3 456 | 13 717 / 13 707 / 13 657 | 3 000 | = | `8b903a` |
| `c5i5` | 7 | 19 530 | 12 464 | 48 533 / 48 521 / 48 461 | 11 755 | = | `dca6a0` |
| `c5i5` | 8 | 97 655 | 45 530 | 169 278 / 169 264 / 169 194 | 44 370 | = (§6.11(g)) | — |
| q5 ($M$ of §6.10) | 6, 7 | 3 905; 19 530 | 3 456; 14 165 | 14 722 / …; 54 097 / 54 085 / 54 025 | 3 214; 13 039 | = | — |
| q5 | 8 | 97 655 | 52 973 | 195 779 / 195 765 / 195 695 | 51 004 | = | — |

("="): the same as the dense run; times are the closure, resp. the whole run of `stdmono.py` including all levels and the summary.) `--sparse` gives the same index, generators and memberships at c4a $n = 5, 6, 7$ and `c5i5` $n = 7$ (local CPU, `results/nondense/cpu_sparse/`). Its form: c4a 17 162 nonzeros at $n = 6$, 101 033 at $n = 7$ (11 per row); `c5i5` $n = 7$ 96 823 (8 per row; the canonical form has 95 107). The C insertion took 1.2 s, 17 s and 7.8 s on a heavily loaded machine (the projection by numpy GEMM took 65–76 min there; on the A10G it is seconds).

`--sparse` at $n = 8$ [computer] (`c5i5`, Modal A10G, 4 CPU and 16 GiB host; `results/nondense/a10g_sparse_n8/`): index 169 278 / 169 264 / 169 194, generators 44 370, free columns $7 = r_8$, the in_D list of §6.11(g) entry for entry; certificate holds. 126 s in all (relators and tests in $A_8$ 22 s; level 8: projection 44.5 s, C insertion 23.8 s; summary with the images modulo $2^{30}, 2^{20}, 2$ 14 s), GPU peak 2.0 GiB, host peak 8.7 GiB (the relators in $A_8$ are 3.1 GB of it). The 129 000 generator rows of level 8 carry 17.0 M nonzeros (relators 9.5 M, torsion shifts 5.9 M, overlaps 1.6 M); the form ends with 573 467 nonzeros, 1.3 % above the canonical 565 950. So at $n = 8$ the bottom-up insertion with tail reduction has essentially no fill-in, as at $n \le 7$.

A by-product [computer]: `c5i5` at $n = 7$ (not measured in §6.11(g)): $w, S, XV, B_{12}, \sum B$, chk $\in \Delta^7$, $B_{13}, B_{23} \notin \Delta^7$, $128X_2X_1^2, 128X_2^2X_1 \in \Delta^7$. With (g): $128X_2^2X_1 \in \Delta^7 \setminus \Delta^8$ — one level above the certified $\Delta^{c+1} = \Delta^6$ (Lemma 10), as conjecture H says (as $x_5$ does at class 4).

**(d) Resources for $d = 6$** *Model of the sizes.* The standard monomials per degree, measured: `c5i5` 5, 25, 125, 580, 2279, 8807, 32842; q5 5, 25, 125, 580, 2497, 9898, 38302 (degrees 1..7), growth factors 4.6 → 3.7. For a six-generator group of the same kind (§6.11(e)3: one more carrier) the factors were taken ×6/5 or +1 (degrees ≤ 3 free), and the top candidates 1.03× the top standard monomials, as measured. Then $C_7 \approx$ 34 000–41 000 (0.61–0.74 $M$) and $C_8 \approx$ 154 000–203 000 (0.46–0.60 $M$). This is a model, not a measurement: the real group may have more or less torsion.

*Costs, scaled from the measured runs* (`c5i5`/q5 at $n = 8$: projection 26.5 s on an A100 and 44.5 s on an A10G, dense Howell 50 s on an A100, sparse insertion 23.8 s, for 45 530 columns and 129 000 generator rows; the projection is $\approx 7 \cdot 10^{12}$ exact multiply-adds per second on the A100; for $d = 6$ it grows ≈ 31×: 3.5× the rows, 9× the work per row):

| | $d = 6, n = 7$ | $d = 6, n = 8$ |
|---|---|---|
| columns (dense $M$) | 34–41 thousand (55 986) | 154–203 thousand (335 922) |
| generator rows of the top level | ≈ 90 000 | ≈ 420 000–500 000 |
| dense form of this method | 2.3–3.4 GB; peak ≲ 25 GB | 48–82 GB; peak several times that: **no** |
| dense method of §6.5 | 6.3 GB; fits an A100 | 226 GB: no |
| projection $\pi_n$ (dense GEMM with $T_{n-1}$) | $\approx 3 \cdot 10^{13}$: seconds | $(5–8) \cdot 10^{15}$: 12–19 min A100, ≈ 23 mi |
| Howell, dense | $\approx 3 \cdot 10^{13}$: 0.5 min A100, 1–2 min A10G; CPU ≈ 7 h | — |
| Howell, `--sparse` | seconds (C, one core) | form 2–4 M nonzeros (13–20 per row, |
| GPU memory, `--sparse` | ≲ 3 GB | ≈ 8–12 GB (2.0 GiB measured at $d =$ 5 |
| host memory | relators in $A_7$: 2 GB | ≈ 20–25 GB (8.7 GiB measured at $d =$ |
| total | **minutes on an A10G** (also CPU, hours) | **`--sparse`: about 25–35 min on one** |

H100: the same memory (80 GB), ≈ 1.5–2× faster. CPU for $n = 8$: the projection by dense GEMM is $\sim 10^{16}$ exact operations (weeks); with sparse $T$ tables (0.6 % nonzero at $n = 6$) it would be hours — not implemented.

**(e) What is left**

1. Done: `--sparse` at $n = 8$ has no fill-in (above). For $d = 6$, $n = 8$ what is missing is the group, not the method.
2. The projection with sparse $T_{n-1}$ (cupyx/scipy CSR): the dominant cost at $d = 6$, $n = 8$, and it would make the CPU route possible.
3. A six-generator group to test (§6.11(e)3): not built; the model of (d) is only as good as the analogy.
4. Storage modulo $2^{16}$ (`uint16`): halves the dense form; the certificate would then be taken at $E = 16$ with $E_2 = 15, 14$, which holds only if the torsion exponent is $\le 2^{14}$ (the largest order in $\Delta/\Delta^2$ of the (3.12)-type groups is $2^{14}$; above that it was not checked).

**Rerun.**

```
# --device numpy on a CPU
python3 gpu/stdmono.py run --input results/c5/presentation/records/c5i5_lin.jsonl.gz --n 6 7 8 --h 1 0 0 0 0
# on Modal
modal deploy gpu/modal_stdmono.py
python gpu/modal_stdmono.py a10g|a100 '<command with $OUT>'
modal volume get math171-nondense <dir> .
```

**Modal spend for §6.13:** $1.69 by `modal billing report` (app `math171-nondense`: the A10G validation at $n \le 7$ with the dense `c5i5` $n = 7$; `c5i5` and q5 at $n = 8$ on an A10G (q5 out of memory there) and on an A100; a statistics run; the first

`--sparse` attempt without the tail reduction, stopped after about 25 min); then $0.09 for `--sparse` at $n = 8$ (one A10G run, 155 s; same app, redeployed). $1.78 in all. All stopped.

## 7. Literature, and whether the method is known

A dagger † marks a reference that was not checked against the source. The DOIs were checked through the Crossref API (title, journal, volume, pages).

- The formula $D_n(F/R) = F \cap (1 + \mathfrak{r} + \mathfrak{f}^n)$ and the Magnus embedding are classics of the theory of dimension subgroups: [MP09] R. Mikhailov, I. B. S. Passi, *Lower Central and Dimension Series of Groups*, Lecture Notes in Math. 1952 (2009), doi:10.1007/978-3-540-85818-8, ch. 1 (the exact section not checked) †; N. Gupta, *Free group rings*, Contemp. Math. 66 (1987), doi:10.1090/conm/066 †. The truncated isomorphism (a) is a special case of Magnus's theorem (W. Magnus, 1937 †).
- [BM23] L. Bartholdi, R. Mikhailov, *Group and Lie algebra filtrations and homotopy groups of spheres*, arXiv:1805.10894 (checked through the arXiv API); journal version J. Topology 16 (2023) 822–853, doi:10.1112/topo.12301 †. Its ancillary file `3group.gap` gives a pc-presentation of a 3-group of order $3^{494}$ on 7 generators with a check of the relations and the class in GAP. There are no computations in the group ring: the proof is theoretical.
- [IMP26] V. Ionin, R. Mikhailov, T. Petrov, *Dimension subrings of Lie rings*, arXiv:2610.02042 (checked through the arXiv API).
- [HMP07] M. Hartl, R. Mikhailov, I. B. S. Passi, *Dimension quotients*, arXiv:0803.3290 (checked through the arXiv API, which gives J. Indian Math. Soc. (N.S.), special volume for the centenary of the IMS (1907–2007), 63–107). It contains no explicit presentation of the Rips group.
- ModIsom (B. Eick, a GAP package): `NilpotentQuotientOfFpAlgebra`, the nilpotent quotient of a finitely presented associative algebra. According to its documentation it works over a finite field (the example and the field `fld` of the result), that is, it gives $\mathbb{F}_p[G]/\Delta^n_{\mathbb{F}_p}$, the Jennings series $M_n$, not the integral $D_n$. Support for $\mathbb{Z}$ was not checked in the sources †. The algorithm: B. Eick, *Computing nilpotent quotients of associative algebras and algebras satisfying a polynomial identity*, IJAC 21 (2011), doi:10.1142/S0218196711006649 †.
- LAGUNA: modular group algebras $\mathbb{F}_p[G]$, again Jennings, not $\mathbb{Z}$.
- The same method was implemented independently (unpublished); the comparison is in §5.

**Conclusion.** The identity $\mathbb{Z}[G]/\Delta^n \cong A_n/J_n$ is classical. Its computational use for the integral $D_n$, that is, linear algebra over $\mathbb{Z}/p^E$ or $\mathbb{Z}$ in $A_n$ of a size that does not depend on $|G|$, we have not found in the literature we searched. The search was not exhaustive.